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Double-sided: tight proofs for guessing games in the quantum random oracle model

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Abstract

The semi-classical One-Way to Hiding (SC-O2H) lemma given by Ambainis et al. (CRYPTO 2019) is a crucial technique to solve the reprogramming problem in the quantum random oracle model (QROM), which can lead to quadratically better bounds for many cases involving guessing games. To achieve tighter bounds, Bindel et al. (TCC, 2019) introduced the double-sided One-Way to Hiding (DS-O2H) lemma, which avoids the loss of query times suffered by the SC-O2H lemma. However, the potential of the DS-O2H lemma to provide better bounds for guessing games has not been considered by far. In this paper, a new double-sided O2H lemma is proposed. By using it, we for the first time give fully tight bounds for several cases involving guessing games. In summary, we show the following results in the QROM: (i) The hardness of inverting a random oracle with the leakage of a one-way injective function can be tightly reduced to the hardness of inverting the involved one-way injective function. (ii) Duman et al. (PKC 2023) introduced the randomness recoverability and defined two transformations ACWC_0 and ACWC relative to random oracles. For ACWC_0 , we prove that its OW-CPA security can be tightly reduced to the OW-CPA security of the underlying public key encryption (PKE) scheme with the randomness recoverability. For ACWC , we design a variant ACWC_1 , and prove that its OW-CPA security can be tightly reduced to the OW-CPA security of the underlying PKE scheme with the unique randomness recoverability (a property slightly stronger than randomness recoverability). (iii) The OW-CPA security of the modular Fujisaki-Okamoto (FO) transformation $\mathbb{T}$ introduced by Hofheinz et al. (TCC 2017), can be tightly reduced to the OW-CPA security of the underlying PKE scheme with the unique randomness recoverability. Additionally, assuming the underlying PKE scheme is unique randomness recoverable, we prove the IND-CCA security of FO-like transformations $\text{FO}_m^\perp$ and $\text{FO}_m^\perp$ and $\text{FO}_m^\perp$ (TCC, 2017) in the QROM, and as far as we know, our proof is tighter than the currently best proof.

Keywords Quantum random oracle model, Public key encryption, Security proof, Fujisaki-Okamoto transformation

Introduction

The random oracle model (ROM) introduced by Bellare and Rogaway (1993) is widely used in the heuristic security analysis of cryptographic schemes utilizing the hash function. This model idealizes the hash function as

a truly random function that can be publicly accessed, i.e. the random oracle. In general, the ROM security proof is more concise than its standard model counterpart. In the quantum setting, an adversary with quantum computational capabilities can compute a hash function in superposition. Based on this fact, Boneh et al. (2011) pointed out that the random oracle model should be lifted to the quantum random oracle model (QROM), where the adversary has quantum access to the random oracle.

Compared with the ROM security, the QROM security of the cryptographic scheme is usually more difficult to achieve, the reason is that some commonly used proof techniques in the ROM can not be directly used in the

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QROM. Indeed, one tricky obstacle we encounter when proving the QROM security is:

- The reprogramming problem. Roughly speaking, to reprogram a random oracle $H : X \rightarrow Y$ at the set $S \subseteq X$ is to replace H with a new random oracle G that satisfies $H(x) = G(x)$ for $x \notin S$. This set S , which contains all the reprogrammed positions, can either be determined at the beginning or be adaptively chosen w.r.t the behavior of the adversary. In the ROM, we claim that an adversary $\mathcal{A}$ can not detect whether H has been reprogrammed into G if $\mathcal{A}$ never queries the random oracle with input from the set S . However, in the QROM, this claim no longer makes sense since $\mathcal{A}$ can query the random oracle with a superposition containing all the possible inputs.

Fortunately, we can solve the above problem (at least in some situations) by using the semi-classical One-Way to Hiding (SC-O2H) lemma introduced by Ambainis et al. (2019). This lemma is a generalization of the original One-Way to Hiding lemma given by Unruh (2015), its crucial novelty is the use of the semi-classical oracle related to the set S .

Semi-classical oracle (informal): For a query to the semi-classical oracle $\mathcal{O}_S^{SC}$ with state $\sum_{x,y} \alpha_{x,y} |x, y\rangle$, $\mathcal{O}_S^{SC}$ first transforms this state into $\sum_{x \in S, y} \alpha_{x,y} |x, y\rangle |1\rangle + \sum_{x \notin S, y} \alpha_{x,y} |x, y\rangle |0\rangle$ and then measures the last qubit. After the measurement, the query state will collapse into the state $\sum_{x \in S, y} \alpha_{x,y} |x, y\rangle$ (non-normalized) if the measurement result is 1, and otherwise $\sum_{x \notin S, y} \alpha_{x,y} |x, y\rangle$ (non-normalized).

Intuitively speaking, the semi-classical oracle $\mathcal{O}_S^{SC}$ performs a binary projective measurement that only checks whether the adversary queries an input from S . Using $\mathcal{O}_S^{SC}$ as a proof technique, the SC-O2H lemma gives an upper bound of the distinguishing probability between random oracles H and G against the adversary $\mathcal{A}$ (i.e. $|\Pr[1 \leftarrow \mathcal{A}^H] - \Pr[1 \leftarrow \mathcal{A}^G]|$) by $O(q) \cdot \sqrt{\text{Adv}(\mathcal{B}^G)}$. Here q is the query times¹ of $\mathcal{A}$ to the random oracle. The simulator $\mathcal{B}$ is an algorithm with oracle access to H , and aims to find an element belonging to the set S . And $\text{Adv}(\mathcal{B}^G)$ denotes the success probability of $\mathcal{B}$.

Although we can use the SC-O2H lemma to solve the reprogramming problem in some situations, the upper bound given by this lemma has a loss related to the query times q and a square-root advantage loss. In (Ambainis et al., 2019, Section 4), they also show that for many

problems involving guessing games (i.e. the game we intend to show that the probability of a certain event is negligible), using their SC-O2H lemma can obtain a tighter proof without the square-root advantage loss (but still has a loss related to the query times q). Roughly speaking, the key point of this tighter proof is the use of the following inequality given in the SC-O2H lemma:

$$\left| \sqrt{\Pr[1 \leftarrow \mathcal{A}^H]} - \sqrt{\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{\mathcal{O}_S^{SC}}]} \right| \leq O(q) \cdot \sqrt{\text{Adv}(\mathcal{B}^G)}.$$

Here Find is the event that $\mathcal{O}_S^{SC}$ ever has measurement result 1 during the running of adversary $\mathcal{A}$ ². When $\mathcal{A}$ is playing a guessing game that searches an element belonging to the set S , without loss of generality, we can assume that $\mathcal{A}$ returns 1 if it successfully finds a value $w \in S$ and always additionally queries its oracle $\mathcal{O}_S^{SC}$ with the state $|w\rangle$. Therefore, by the definition of $\mathcal{O}_S^{SC}$ shown above and the assumption of $\mathcal{A}$, it is impossible that $\mathcal{A}$ returns 1 and $\neg \text{Find}$ occurs simultaneously. Hence, for this type of adversary, the probability $\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{\mathcal{O}_S^{SC}}]$ must always be zero, and then we bound $\Pr[1 \leftarrow \mathcal{A}^H]$ by $O(q^2) \cdot \text{Adv}(\mathcal{B}^G)$ without the square-root advantage loss.

Later, at the cost of limited applicability, Bindel et al. (2019) propose the double-sided O2H (DS-O2H) lemma without incurring the loss related to the query times. This lemma gives an upper bound of the distinguishing probability between random oracles H and G against the adversary $\mathcal{A}$ (i.e. $|\Pr[1 \leftarrow \mathcal{A}^H] - \Pr[1 \leftarrow \mathcal{A}^G]|$) by $\sqrt{\text{Adv}(\mathcal{C}^{H,G})}$, where $\text{Adv}(\mathcal{C}^{H,G})$ denotes the success probability of $\mathcal{C}$, and the simulator $\mathcal{C}$ is an algorithm that aims to find an element belonging to the set S ³ with oracle access to H and G . In addition, the DS-O2H lemma also upper bounded $|\sqrt{\Pr[1 \leftarrow \mathcal{A}^H]} - \sqrt{\Pr[1 \leftarrow \mathcal{A}^G]}|$ by $\sqrt{\text{Adv}(\mathcal{C}^{H,G})}$.

Compared with the SC-O2H lemma, the DS-O2H lemma has a tighter upper bound but requires that the adversary $\mathcal{C}$ can make queries to both H and G , which becomes a limit of the applicability. One may curious about that whether the DS-O2H lemma can be applied to some problems involving guessing games to obtain a tight bound, since its also bounds $|\sqrt{\Pr[1 \leftarrow \mathcal{A}^H]} - \sqrt{\Pr[1 \leftarrow \mathcal{A}^G]}|$ by $\sqrt{\text{Adv}(\mathcal{C}^{H,G})}$ and avoids the loss related to the query times. However, because the semi-classical oracle $\mathcal{O}_S^{SC}$ is not introduced in

¹ Ambainis et al. (2019) consider the parallel query (as well as our proof in later sections), and q actually denotes the query depth of $\mathcal{A}$. In this introduction, we view q as the total query times of $\mathcal{A}$ for simplicity.

² The adversary $\mathcal{A}$ involved in the probability $\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{\mathcal{O}_S^{SC}}]$ actually queries $H \setminus S$, where $H \setminus S$ is an oracle that first queries $\mathcal{O}_S^{SC}$ and then H . Here we directly write $\mathcal{O}_S^{SC}$ for notational simplicity.

³ Bindel et al. (2019) only consider S satisfying $|S| = 1$.

the DS-O2H lemma, it is difficult to adjust the structure of $\mathcal{A}$ to make $\Pr[1 \leftarrow \mathcal{A}^G] = 0^4$. This is because, in the DS-O2H lemma, $\mathcal{A}$ is only an adversary that distinguishes H and G , and it is challenging to achieve a significant distinguishing advantage by adjusting the structure of $\mathcal{A}$. Therefore, by using the DS-O2H lemma, we can only obtain $\sqrt{\Pr[1 \leftarrow \mathcal{A}^H]} \leq \sqrt{\Pr[1 \leftarrow \mathcal{A}^G]} + \sqrt{\text{Adv}(\mathcal{C}^{H,G})}$ and can not obtain $\Pr[1 \leftarrow \mathcal{A}^H] \leq \text{Adv}(\mathcal{C}^{H,G})$.

Overall, for many problems involving guessing games (at least for problems considered in (Ambainis et al., 2019, Section 4)), we can use the SC-O2H lemma to obtain a proof without the square-root advantage loss but suffering from the loss related to the query times. In contrast, the DS-O2H lemma incurs a square-root advantage loss, but it avoids the loss related to the query times by giving the simulator one more oracle access. Hence, for many problems involving guessing games, there exists a natural question:

*Is it possible to propose a new O2H lemma that combines the SC-O2H lemma and the DS-O2H lemma, and use it to give a **tight** proof?*

Here a tight proof avoids both the loss related to the query times and the square-root advantage loss.

Our contributions

A new O2H lemma: To answer the above question, we prove the following new O2H lemma that might be of independent interest, and it can be viewed as a combination of the SC-O2H lemma and the DS-O2H lemma.

Lemma 1 (Double-sided O2H, with semi-classical oracle, informal⁵) *For a random function $H : X \rightarrow Y$ and a set $S := \{w^*\}$ with only one element $w^* \in X$, let $\mathcal{A}^H(z)$ be a quantum oracle algorithm. (H, G, S, z may have arbitrary joint distribution) Then, there exists a quantum oracle algorithm $\mathcal{B}(z)$ such that*

$$\left| \sqrt{\Pr[1 \leftarrow \mathcal{A}^H(z)]} - \sqrt{\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)]} \right| \leq 2\sqrt{P_{\text{extract}}}.$$

Here Find is the event that $\mathcal{A}$ ever queries the semi-classical oracle $\mathcal{O}_S^{\text{SC}}$ with an input state $|w^*\rangle$, and P_{extract} is $\Pr[w = w^* : w \leftarrow \mathcal{B}^{H, H \setminus S}(z)]$, where $H \setminus S$ is the oracle that first queries the semi-classical oracle $\mathcal{O}_S^{\text{SC}}$ and then H .

The above lemma has the same requirement as the DS-O2H lemma, i.e. H and G can only be different at one

⁴ Note that in the SC-O2H lemma, $\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{\mathcal{O}_S^{\text{SC}}}]$ can be 0 by directly adjusting the structure of $\mathcal{A}$.

⁵ The formal version of this lemma is stated as Lemma 4

point and the algorithm $\mathcal{B}$ has access to two oracles. The difference from the DS-O2H lemma is that our lemma introduces the semi-classical oracle and requires the algorithm $\mathcal{B}$ to have the oracle access to both H and $H \setminus S$.

By using our new O2H lemma, we present tight proofs for the following problems that involve the guessing game:

Inverting a random oracle with leakage: Consider the following problem: Let $H : X \rightarrow Y$ be a random oracle and P be a one-way injective function with domain X . Then for $w^* \xleftarrow{\$} X$, how hard is it to invert $H(w^*)$ with the leakage $P(w^*)$? This problem is a variant of the problem considered in Section 4.2 of Ambainis et al. (2019), where their assumption is that distributions $\{(x, P(x)) : x \xleftarrow{\$} X\}$ and $\{(x_1, P(x_2)) : x_1, x_2 \xleftarrow{\$} X\}$ are hard to distinguish.

However, their reduction is not tight since it has a loss related to the query times of H . In this paper, assuming P is a one-way injective function instead, we give a tight reduction from the hardness of inverting $H(w^*)$ with the leakage $P(w^*)$ to the hardness of inverting $P(w^*)$, in the QROM.

QROM security proof of ACWC₀: The ROM transformation ACWC₀ is introduced by Duman et al. (2023) and used to transform a δ -average-correct public key encryption (PKE) scheme into a δ' -worst-correct PKE scheme. As proved by Duman et al. (2023), ACWC₀ can transform a PRE-CPA secure and randomness recoverable PKE scheme into an IND-CPA secure PKE scheme, in the QROM.

Here a randomness recoverable PKE scheme satisfies that there exists an efficient algorithm Rec such that for a plaintext m and ciphertext c , $\text{Rec}_{pk}(m, c) = r$ and $\text{Enc}_{pk}(m; r) = c$. That is, for an input (m, c) , Rec will recover a suitable randomness r that makes the encryption of m to be c . In the following, we also call Rec the recover algorithm.

However, as shown in Table. 1, the QROM security proof of ACWC₀ given by Duman et al. (2023) is not tight and has a square-root advantage loss. In this paper, we prove that ACWC₀ tightly transforms a OW-CPA secure and unique randomness recoverable PKE scheme into a OW-CPA secure PKE scheme, in the QROM.

QROM security proof of ACWC_†: Duman et al. (2023) also introduced a ROM transformation ACWC_† that transforms a δ -average-correct PKE scheme into a δ' -worst-correct PKE scheme. Compared with ACWC₀, ACWC_† has the advantage that it avoids adding the length of the ciphertext.

Table 1 Security bounds of different transformations in the QROM.

Transformation	Underlying security	Requirement	Achieved security	Security bound
ACWC ₀ (Duman et al. 2023)	PRE-CPA	RR	IND-CPA	$d\sqrt{\epsilon}$
ACWC ₀ Our work	OW-CPA	URR	OW-CPA	$\epsilon + \delta_a$
ACWC (Duman et al. 2023)	PRE-CPA	RR	OW-CPA	$q^2 \cdot \epsilon$
ACWC ₁ Our work	OW-CPA	URR	OW-CPA	$\epsilon + \delta_a$
T (Bindel et al. 2019)	IND-CPA	\	OW-CPA	$d \cdot \epsilon$
T Our work	OW-CPA	URR	OW-CPA	$\epsilon + \delta_a$
FO _m ^L (Jiang et al. 2018)	OW-CPA	\	IND-CCA	$q\sqrt{\epsilon} + q\sqrt{\delta_w}$
FO _m ^L (Kuchta et al. 2020)	IND-CPA	η -injective	IND-CCA	$d^2 \cdot \epsilon + dq \cdot \delta_w + q \cdot \sqrt{\eta}$
FO _m ^L (Hövelmanns and Majenz 2023)	IND-CPA	γ -spread	IND-CCA	$4\sqrt{(d + q_D) \cdot \epsilon} + q^2 \cdot \delta_w + q_D(q + q_D)\sqrt{2^{-\gamma}}$
FO _m ^L (Ge et al. 2023)	IND-CPA	γ -spread	IND-CCA	$d(d + q_D) \cdot \epsilon + q \cdot \sqrt{\delta_w} + q_D\sqrt{2^{-\gamma}}$
FO _m ^L , FO _m ^L Our work	OW-CPA	URR	IND-CCA	$(d + q_D) \cdot \epsilon + q \cdot \sqrt{\delta_w}$

Here q is the total number of query times to random oracles, d is the query depth of random oracles, and q_D is the adversary's query times to the decapsulation oracle. δ_a , δ_w and ϵ are the average correctness error, the worst correctness error and the security bound of the underlying PKE scheme, respectively. In this table, we abbreviate randomness recoverable as RR and unique randomness recoverable as URR for simplicity

(1) The detailed definitions of the security notions, the requirement written in this table, and the average/worst correctness error can be found in Appendix B

(2) Actually, Ge et al. (2023) prove the IND-qCCA security of FO_m^L. Since IND-qCCA security implies IND-CCA security, we write their achieved security as IND-CCA for notional simplicity

(3) In the security proof of ACWC₀ and ACWC given by Duman et al. (2023), both the plaintext and the encryption randomness of the underlying rPKE scheme can be sampled from some special distributions (perhaps not uniform). Although in our security proofs of ACWC₀ and ACWC₁, we suppose both the plaintext and the encryption randomness of the underlying randomized PKE scheme are sampled from the uniform distribution for convenience, we stress that our proof can be extended to the non-uniform case in a straightforward manner

(4) According to the definitions (given in Appendix B) of these above properties written in the "Requirement" column, URR is a slightly stronger requirement compared to RR and γ -spread

However, as shown in Table. 1, the security proof of ACWC given by Duman et al. (2023) is not tight in the QROM. In this paper, we design a new ROM transformation ACWC₁, a variant of ACWC. ACWC₁ has the same function as ACWC and also avoids adding the length of the ciphertext. In the QROM, we prove that ACWC₁ tightly transforms a OW-CPA secure and unique randomness recoverable PKE scheme into a OW-CPA secure PKE scheme.

Here the requirement of unique randomness recoverability is slightly stronger than the randomness recoverability,⁶ it additionally requires that if $\text{Enc}_{pk}(m; r) = c$, then $\text{Rec}_{pk}(m, c) = r$. That is, for an input $(m, c = \text{Enc}_{pk}(m; r))$, the recover algorithm Rec of a unique randomness recoverable PKE will directly recover the randomness r that used to encrypt m to obtain c .

QROM security proof of T: The modular Fujisaki-Okamoto (FO) transformation T introduced by Hofheinz et al. (2017) is a ROM transformation that transforms a randomized public key encryption (rPKE) scheme into a deterministic public key encryption (dPKE) scheme. The tightest security proof of T in the QROM as far as we

know is given by Bindel et al. (2019), which shows that T can transform an IND-CPA secure rPKE scheme into a OW-CPA secure dPKE scheme.

However, as shown in Table. 1, the QROM security proof of T given by Hofheinz et al. (2017) is non-tight. In this paper, we prove that in the QROM, T tightly transforms a OW-CPA secure and unique randomness recoverable rPKE scheme into a OW-CPA secure dPKE scheme.

Besides the above tight proofs of some problems involving the guessing game, for the FO-like transformations FO_m^L and FO_m^L introduced by Hofheinz et al. (2017), which are used to transform a rPKE scheme into a key encapsulation mechanism (KEM) scheme, we obtain the following results:

- In the QROM, with the requirement of unique randomness recoverability, we prove that the IND-CCA security of FO_m^L and FO_m^L can both be reduced to the OW-CPA security of the corresponding underlying rPKE scheme. The detailed bounds are given in Table. 1. Compared with the tightest security proof of FO_m^L (resp. FO_m^L) by far, which is given by Kuchta et al. (2020) (resp. Ge et al. (2023)), our QROM security proof of FO_m^L (resp. FO_m^L) is much tighter and

⁶ We note that the generic NTRU-based schemes designed by Duman et al. (2023) are also unique randomness recoverable.

only requires the underlying rPKE scheme to be OW-CPA (rather than IND-CPA) secure.

Technique overview

At the heart of our results is the new O2H lemma (i.e. Lemma 1). In this section, we first give a proof sketch of Lemma 1, and then we explain how to use this lemma to obtain the tight proofs shown in our work. Finally, we show how to give the QROM security proofs of $\text{FO}_m^\perp$ (resp. $\text{FO}_m^\perp$).

Proof sketch of Lemma 1: We first consider fixed (H, G, S, z) and suppose the algorithm $\mathcal{A}$ in Lemma 1 is a unitary oracle algorithm.⁷ We also suppose that $\mathcal{A}$ acts on the quantum register Q and makes q oracle queries. Let l_i ($i = 1, \dots, q$) be a one-qubit quantum register with the initial state $|0\rangle$. Following the proof of the SC-O2H lemma given by Ambainis et al. (2019), the algorithm $\mathcal{A}^{H \setminus S}$ can be purified into a unitary oracle algorithm $\mathcal{A}_{uni}^H$ that acts on registers $Q, l_1, \dots, l_q$. Roughly speaking, the core idea is that the binary measurements performed by $\mathcal{O}_S^{SC}$ in algorithm $\mathcal{A}^{H \setminus S}$ can be deferred and are performed by measuring registers $l_1, \dots, l_q$ all at once just after the final measurement on register Q .

Then, with the same algorithm $\mathcal{B}$ designed in the proof of the DS-O2H lemma given by Bindel et al. (2019), we find $P_{\text{extract}} := \Pr[w = w^* : w \leftarrow \mathcal{B}^{H, H \setminus S}(z)]$ satisfies

$$B(|\Psi_H\rangle_Q |0\rangle_{l_1} \dots |0\rangle_{l_q}, |\Psi_{H \setminus S}\rangle_{Q, l_1, \dots, l_q}) \leq 2\sqrt{P_{\text{extract}}}.$$

Here, if the state $|\psi\rangle$ is on the quantum register R , we denote it as $|\psi\rangle_R$. The state $|\Psi_H\rangle_Q$ (resp. $|\Psi_{H \setminus S}\rangle_{Q, l_1, \dots, l_q}$) denotes the state of $\mathcal{A}^H(z)$ (resp. $\mathcal{A}_{uni}^H(z)$) before the final measurement. $B(|\psi_1\rangle, |\psi_2\rangle)$ denotes the Bures distance in Nielsen and Chuang (2016) between two density operators $|\psi_1\rangle\langle\psi_1|$ and $|\psi_2\rangle\langle\psi_2|$. Next, by designing a special binary measurement and applying Lemma 3 and Lemma 4 in Ambainis et al. (2019), we obtain

$$|P_{\text{left}} - P_{\text{right}}| \leq B(|\Psi_H\rangle_Q |0\rangle_{l_1} \dots |0\rangle_{l_q}, |\Psi_{H \setminus S}\rangle_{Q, l_1, \dots, l_q}),$$

where $P_{\text{left}} := \sqrt{\Pr[1 \leftarrow \mathcal{A}^H(z)]}$ and

$$P_{\text{right}} := \sqrt{\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)]}.$$

Finally, for fixed (H, G, S, z) and the unitary oracle algorithm $\mathcal{A}$, Lemma 1 is proved by combining the above two inequalities. Then, for (H, G, S, z) sampled from arbitrary joint distribution and non-unitary oracle algorithm $\mathcal{A}$, we can follow the above proof idea to prove Lemma 1, where the only difference is that we need to construct a unitary variant of $\mathcal{A}$ at first and the final states of $\mathcal{A}^H$ and

$\mathcal{A}_{uni}^H$ are mixed states. The detailed proof of Lemma 1 can be found in Appendix A.

How to obtain the tight proofs shown in our work: In Sect. 1.1, we list the following results obtained by using our Lemma 1:

- For a quantum random oracle $H : X \rightarrow Y$, and a one-way injective function P with domain X , the hardness of inverting $H(w^*)$ with the leakage $P(w^*)$ can be tightly reduced to the hardness of inverting $P(w^*)$.
- Tight security proofs of transformations ACWC_0 , ACWC_1 and T in the QROM.

Indeed, the above tight proofs are obtained similarly. Here we particularly explain how we tightly reduce the hardness of inverting $H(w^*)$ with the leakage $P(w^*)$ to the hardness of inverting $P(w^*)$. Then, we will point out the difference between this proof and the other tight security proofs of ACWC_0 , ACWC_1 and T in the QROM.

For $w^* \xleftarrow{\$} X$, let $\mathcal{A}^H(H(w^*), P(w^*))$ be an algorithm that intends to invert $H(w^*)$. Denote the classical output of $\mathcal{A}^H(H(w^*), P(w^*))$ as w and define

$$P_{\text{left}} := \Pr[w = w^* : w \leftarrow \mathcal{A}^H(H(w^*), P(w^*))].$$

Roughly speaking, our tight proof consists of the following three steps:

1. Design a new algorithm $\mathcal{A}_1$: Let $\mathcal{A}_1^H(H(w^*), P(w^*))$ be an algorithm that is identical to $\mathcal{A}^H(H(w^*), P(w^*))$ except that it makes an additional query to H on the state $|w\rangle$ (w is the output of $\mathcal{A}^H(H(w^*), P(w^*))$) and finally outputs a bit $\llbracket w = w^* \rrbracket$. Here $\llbracket w = w^* \rrbracket$ equals 1 if $w = w^*$ and otherwise 0. Obviously, we have

$$P_{\text{left}} = \Pr[1 \leftarrow \mathcal{A}_1^H(H(w^*), P(w^*))].$$

2. Use our new O2H lemma: Define a set $S := \{w^*\}$, then by using Lemma 1 with the algorithm $\mathcal{A}_1$ and the set S , there exists an algorithm $\mathcal{B}$ such that

$$|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}| \leq 2\sqrt{\Pr[w^* \leftarrow \mathcal{B}^{H, H \setminus S}(H(w^*), P(w^*))]}A.$$

Here $P_{\text{right}} := \Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}_1^{H \setminus S}(H(w^*), P(w^*))]$. Note that $H \setminus S$ is an oracle that first queries the semi-classical oracle $\mathcal{O}_S^{SC}$ and then H , Find is the event that $\mathcal{O}_S^{SC}$ is ever queried with an input state $|w^*\rangle$. Hence, according to the definition of $\mathcal{A}_1$, it is impossible for events $b = 1$ and $\neg \text{Find}$ to occur simultaneously. This means that $P_{\text{right}} = 0$ and thus we obtain

$$P_{\text{left}} \leq 4 \Pr[w^* \leftarrow \mathcal{B}^{H, H \setminus S}(H(w^*), P(w^*))].$$

⁷ That is, an algorithm which starts with a pure state, alternately applies a fixed unitary operation and performs oracle queries, and finally performs a projective measurement on the final state to get the classical output.

3. Simulate H and $H \setminus S$ for $\mathcal{B}$: By the definition of the semi-classical oracle $\mathcal{O}_S^{SC}$, if we can recognize whether an element is in the set S , we can implement this oracle. In our setting, we let $S := \{w^*\}$, and thus $x \in S$ iff $x = w^*$. Since P is an injective function, we can recognize whether x is in the set S by computing $P(x)$. That is, if $P(x) = P(w^*)$, then we have $x \in S$, otherwise $x \notin S$. Then, for $w^* \xleftarrow{\$} X$, we construct the following algorithm $\mathcal{C}$ that takes $P(w^*)$ as the input and aims to invert it:

- Choose a function H^* to simulate a quantum random oracle⁸ with input/output space X/Y . Then choose $z \xleftarrow{\$} Y$, and invoke $\mathcal{B}$ with the input $(z, P(w^*))$. Finally, return the output of $\mathcal{B}$.
- When $\mathcal{B}$ queries H with the state $|x_H, y_H\rangle$, if $P(x_H) = P(w^*)$ (i.e. $x_H = w^*$ since P is injective), return $|x_H, y_H + z\rangle$, otherwise return $|x_H, y_H + H^*(x_H)\rangle$, where $+: Y \times Y \rightarrow Y$ is a group operation on Y .
- When $\mathcal{B}$ queries $H \setminus S$ with the state $|x_H, y_H\rangle$, first use $P(w^*)$ to implement the oracle $\mathcal{O}_S^{SC}$, then perform the operation used in answering queries to H (i.e. (b)).

It is obvious that $\mathcal{C}(P(w^*))$ perfectly simulates the environment of $\mathcal{B}$, hence

$$\begin{aligned} \Pr[w = w^* : w \leftarrow \mathcal{C}(P(w^*))] \\ = \Pr[w = w^* : w \leftarrow \mathcal{B}^{H, H \setminus S}(H(w^*), P(w^*))]. \end{aligned}$$

Combining the formulas shown in the above three steps, we conclude that for $w^* \xleftarrow{\$} X$,

$$\begin{aligned} \Pr[w = w^* : w \leftarrow \mathcal{A}^H(H(w^*), P(w^*))] \\ \leq 4 \Pr[w = w^* : w \leftarrow \mathcal{C}(P(w^*))]. \end{aligned}$$

That is to say, the hardness of inverting $H(w^*)$ with the leakage $P(w^*)$ can be tightly reduced to the hardness of inverting $P(w^*)$.

The security proofs of transformations ACWC₀, ACWC₁, and T in the QROM also follow the above three steps. However, since the “injective function P ” no longer exists, in the step 3., we need to use alternative methods to simulate the oracles H and $H \setminus S$ for $\mathcal{B}$, which is the main difference compared to the above proof.

Roughly speaking, in the QROM security proofs of ACWC₀, ACWC₁ and T, the input of $\mathcal{B}$ is $c^* := \text{Enc}_{pk}(m^*; H(m^*))$ ⁹ and the set S is $\{m^*\}$. Here

Enc is the encryption algorithm of the underlying PKE scheme, and m^* is uniformly sampled from the message space $\mathcal{M}$. $H(m^*)$ is the randomness used to encrypt m^* , which is uniformly random in the randomness space $\mathcal{R}$ since H is a random oracle. As mentioned above, by the definition of the semi-classical oracle $\mathcal{O}_S^{SC}$, the key point to simulate the oracles H and $H \setminus S$ is the ability to recognize whether an element is in the set S (i.e. whether an element is m^*). To accomplish this recognition, in our proof, we first prove the following lemma¹⁰:

Lemma 2 For a δ -average/worst-correct PKE scheme (Gen, Enc, Dec) with message space $\mathcal{M}$ and randomness space $\mathcal{R}$, we have

$$\Pr \left[\begin{array}{c} \exists (m' \neq m^*, r') \in \mathcal{M} \times \mathcal{R}, \\ \text{Enc}_{pk}(m'; r') = \text{Enc}_{pk}(m^*; r^*) \end{array} \right] \leq 2\delta,$$

where the probability is taken over $(pk, sk) \leftarrow \text{Gen}$, $m^* \xleftarrow{\$} \mathcal{M}$, and $r^* \xleftarrow{\$} \mathcal{R}$.

This lemma shows that, for $\mathcal{B}$'s input $c^* := \text{Enc}_{pk}(m^*; H(m^*))$, if there exists a pair (m', r') satisfying $\text{Enc}_{pk}(m'; r') = c^*$, then $m' = m^*$ holds with the error term 2δ . Thus it seems that we can use this lemma to help us recognize m^* .

However, the input space of the quantum random oracle H only involves $\mathcal{M}$, so when $\mathcal{B}$ queries H , we can only get m' and r' always remains unknown. In order to overcome this obstacle, we suppose that the underlying PKE scheme also satisfies the unique randomness recoverability,¹¹ which means that there exists an algorithm Rec to recover the randomness used in the encryption process given the corresponding plaintext and ciphertext as input. Therefore, we have $\text{Rec}_{pk}(m^*, c^*) = H(m^*)$.

Now, with the assistance of Lemma 2 and the recover algorithm Rec, we can identify whether m' is m^* as follows:

- Compute $\text{Rec}_{pk}(m', c^*)$, and if the equation $\text{Enc}_{pk}(m'; \text{Rec}_{pk}(m', c^*)) = c^*$ holds, we regard m' as the element m^* .

Obviously, if $m' = m^*$, the above method can successfully identify m' as m^* . If $m' \neq m^*$, then regardless of the value $\text{Rec}_{pk}(m', c^*)$, the probability of identifying m' as m^* will not exceed 2δ by Lemma 2.

How to give the QROM security proofs of $\text{FO}_m^\perp$ and $\text{FO}_m^\perp$: Here we only focus on $\text{FO}_m^\perp$, since Theorem 3 in Bindel

⁸ The method for choosing H^* is shown in Sect. 2.2.

⁹ The definition of c^* in ACWC₀/ACWC₁ is slightly different. Without loss of generality, we write c^* as $\text{Enc}_{pk}(m^*; H(m^*))$ in this technique overview.

¹⁰ This lemma can also be viewed as a general version of Lemma 4 given by Liu and Wang (2021).

¹¹ The security proof of ACWC₀ only requires a weaker property named randomness recoverability.

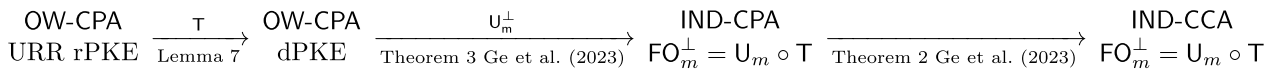


Fig. 1 Proof overview of $\text{FO}_m^\perp$ in the QROM. Here we abbreviate unique randomness recoverable as URR for simplicity

et al. (2019) demonstrates that the security of $\text{FO}_m^\perp$ implies the security $\text{FO}_m^\perp$. Recall that Hofheinz et al. (2017) have defined $\text{FO}_m^\perp$ as $\text{U}_m^\perp \circ \text{T}$, where $\text{U}_m^\perp$ is a ROM transformation that transforms a dPKE scheme into a KEM scheme.

In Fig. 1, we give a proof outline of $\text{FO}_m^\perp$ in the QROM. The Theorem 2 given in Ge et al. (2023) shows that the IND-CCA security of $\text{FO}_m^\perp$ can be reduced to its IND-CPA security. Moreover, the Theorem 3 given in Ge et al. (2023) shows that the IND-CPA security of $\text{U}_m^\perp$ can be reduced to the OW-CPA security of the underlying dPKE scheme. Therefore, by combining these two theorems with our Lemma 7, we obtain an IND-CCA security proof of $\text{FO}_m^\perp$.

Related work

Kuchta et al. (2020) also give a new double-sided O2H lemma named measure-rewind-measure O2H (MRM-O2H). Different from the SC-O2H lemma and the DS-O2H lemma, their new lemma avoids the square-root advantage loss for the first time but still incurs a loss related to the query times, resulting in a non-tight bound.

There are also some O2H variants that combine the SC-O2H lemma with the compressed oracle technique given by Zhandry (2019), including examples in Czajkowski et al. (2019); Ge et al. (2023); Katsumata et al. (2020). From an application perspective, our new O2H lemma and those O2H variants in Czajkowski et al. (2019); Ge et al. (2023); Katsumata et al. (2020) have different application scenarios. For example, the Compressed Semi-Classical One-Way to Hidding (CSC-O2H) lemma given by Ge et al. (2023), its application is to substitute a unitary operation “ Ext_{f_1} ” with another one “ Ext_{f_2} ” under the condition of the measurement result of “compressed semi-classical oracle” is never 1, while the application of our new O2H lemma in this paper, as explained in the technique outline, is to make the adversary’s success probability be 0 under the condition of the measurement result of semi-classical oracle $\mathcal{O}_S^{\text{SC}}$ is never 1.

According to the definition of the unique randomness recoverability given in Appendix B, this property is only a requirement of the underlying rPKE scheme rather than a security assumption. Moreover, if a rPKE scheme is OW-CPA secure and unique randomness recoverable, then it cannot be IND-CPA secure. As explained in the technique overview, we can use Lemma 2 to construct an

algorithm to distinguish which plaintext is encrypted by the challenger. So it seems that the OW-CPA security plus the unique randomness recoverability is a simpler cryptanalytic target than the IND-CPA security. Meanwhile, many NTRU-based rPKE schemes proposed recently satisfy the unique randomness recoverability. For example, the GenNTRU[η] proposed by Duman et al. (2023), the GenNTRU[ψ_1^n] proposed by Kim and Park (2023) and the NEV-PKE’ proposed by Zhang et al. (2023).

Preliminaries

Notation

Let $\llbracket x = y \rrbracket$ be a bit that equals 1 if $x = y$ and otherwise 0. Denote $H : X \rightarrow Y$ as a function with domain X and codomain Y , Ω_H as a set that contains all those functions H . We use $x \xleftarrow{\$} S$ to denote that x is a uniformly random element sampled from a finite set S . We also use $x \leftarrow \mathcal{D}$ to denote that the chosen x is subject to distribution $\mathcal{D}$. $\Pr[A : B]$ is the probability that the predicate A keeps true where all variables in A are conditioned according to predicate B . For a function or algorithm $\mathcal{A}$, $\text{Time}(\mathcal{A})$ denotes the time complexity of (an algorithm computing) $\mathcal{A}$.

Quantum random oracle model

We briefly describe the quantum random oracle model (QROM) introduced by Boneh et al. (2011). In the random oracle model (ROM), a uniformly random function $H : X \rightarrow Y$ is selected as a random oracle, and all parties have access to H . In the quantum setting, the QROM is considered and the adversary has quantum access to the random oracle H in this model. In the QROM, we take the random oracle H as a unitary operation $O_H : |x_H, y_H\rangle \mapsto |x_H, y_H \oplus H(x)\rangle$ and also call H a quantum random oracle (QRO).

Simulation of QRO. Let t_{RO} denote the time needed to simulate one query to the QRO. As summarized by Xagawa and Yamakawa (2019), there exist three different methods to simulate a QRO, each has its own strong and weak points.

1. The first method uses a $2q$ -wise independent hash function. For an adversary $\mathcal{A}$ makes at most q queries to its oracle, Zhandry (2012) proved that the real QRO is perfectly indistinguishable from an oracle simulated by a $2q$ -wise independent hash function.

The drawback of this method is that its t_{RO} is $O(q)$, which will cause a running time blowup of a reduction algorithm.

2. The second method uses a quantum secure PRF, which is first used by Boneh et al. (2011). The t_{RO} of this method is the time needed for evaluating a PRF, which is usually much smaller than $O(q)$. The drawback of this method is that it needs an assumption that the quantum secure one-way function exists (Zhandry 2012).
3. The third method uses a real hash function like SHA-2. The t_{RO} of this method is the time needed for evaluating a hash function, which is usually much smaller than the above two methods. The drawback of this method is that it even needs a stronger assumption than the existence of quantum secure one-way functions.

Lemma 3 (Double-sided O2H; Bindel et al. (2019) Lemma 5) *Let $H, G : X \rightarrow Y$ be random functions, let z be a random value, and let $S \subset X$ be a random set such that $\forall x \notin S, H(x) = G(x)$. (H, G, S, z may have arbitrary joint distribution.) Let $\mathcal{A}^H$ be a quantum oracle algorithm. Let $f : X \rightarrow W \subseteq \{0, 1\}^n$ be any function, and let $f(S)$ denote the image of S under f . Let Ev be an arbitrary classical event.*

We will define another quantum oracle algorithm $\mathcal{B}^{H,G}(z)$. This $\mathcal{B}$ runs in about the same amount of time as $\mathcal{A}$, but when $\mathcal{A}$ queries H, B queries both H and G , and also runs f twice. Let

$$P_{\text{left}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^H(z)], \quad (1a)$$

$$P_{\text{right}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^G(z)], \quad (1b)$$

$$P_{\text{extract}} := \Pr[w \in f(S) : w \leftarrow \mathcal{B}^{H,G}(z)] \quad (1c)$$

If $f(S) = \{w^\}$ is a single element, then $\mathcal{B}$ will only return $\perp$ or w^* , and furthermore*

$$|P_{\text{left}} - P_{\text{right}}| \leq 2\sqrt{P_{\text{extract}}}$$

and

$$|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}| \leq 2\sqrt{P_{\text{extract}}}.$$

Semi-classical oracle Ambainis et al. (2019). For a subset $S \subset X$, let $f_S : X \rightarrow \{0, 1\}$ be a function that $f_S(x) = 1$ iff $x \in S$. The semi-classical oracle $\mathcal{O}_S^{SC}$ performs the following operations on the input state $\sum_{x \in X, z \in Y} \alpha_{x,z} |x, z\rangle$:

1. Initialize a single qubit B with $|0\rangle$
2. Transform state $\sum_{x \in X, z \in Y} \alpha_{x,z} |x, z\rangle |0\rangle$ into $\sum_{x \in X, z \in Y} \alpha_{x,z} |x, z\rangle |f_S(x)\rangle$.
3. Measure B and output the measurement outcome.

By the above definition, one can implement the semi-classical oracle $\mathcal{O}_S^{SC}$ within time $\text{Time}(f_S)$ if it can compute the function f_S . In the execution of an algorithm that has access to $\mathcal{O}_S^{SC}$, we denote Find as the event that $\mathcal{O}_S^{SC}$ ever outputs 1. For a random oracle $H : X \rightarrow Y$, we also define $H \setminus S$ (" H punctured on S ") as an oracle which first queries $\mathcal{O}_S^{SC}$ and then H .

Now we give a new double-sided O2H lemma as follows, which can be viewed as an adapted version of Lemma 3. It introduces the semi-classical oracle and bounds the difference of the square-root of adversary probabilities between two games, where one game considers H and the other considers $H \setminus S$ or $G \setminus S$. The proof of the following lemma is similar to the proof of the semi-classical O2H lemma (Ambainis et al. 2019), and we give it in Appendix A.

Lemma 4 (Double-sided O2H, with semi-classical oracle) *Let $H, G : X \rightarrow Y$ be random functions, let z be a random value, and let $S \subset X$ be a random set such that $\forall x \notin S, H(x) = G(x)$. (H, G, S, z may have arbitrary joint distribution.) Let $\mathcal{A}^H$ be a quantum oracle algorithm of query times q . Let $f : X \rightarrow W \subseteq \{0, 1\}^n$ be any function, let $f(S)$ denote the image of S under f .*

We will define a quantum oracle algorithm $\mathcal{B}(z)$ with access to H and G (or H and $H \setminus S$) running in $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q) \cdot \text{Time}(f)$,¹² let

$$P_{\text{left}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^H(z)],$$

$$P_{\text{right}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^G(z)], \quad (\text{Case 1})$$

$$P_{\text{extract}} := \Pr[w \in f(S) : w \leftarrow \mathcal{B}^{H,G}(z)],$$

$$P_{\text{extract}}' := \Pr[w \in f(S) : w \leftarrow \mathcal{B}^{H,H \setminus S}(z)].$$

If $f(S) = \{w^\}$ is a single element, then $\mathcal{B}$ will only return $\perp$ or w^* . Furthermore, we have*

$$|P_{\text{left}} - P_{\text{right}}| \leq 2\sqrt{P_{\text{extract}}}$$

and

$$|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}| \leq 2\sqrt{P_{\text{extract}}}.$$

¹² If f is an identical function such that $f(x) = x$ for any $x \in X$, then we can omit the term $\text{Time}(f)$ here, since it can be ignored under this definition of f .

The lemma also holds with bound $2\sqrt{P_{\text{extract}}'}$ for the following alternative definitions of P_{right} :

$$P_{\text{right}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^{H \setminus S}(z)], \quad (\text{Case 2})$$

$$P_{\text{right}} := \Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)], \quad (\text{Case 3})$$

$$P_{\text{right}} := \Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{G \setminus S}(z)], \quad (\text{Case 4})$$

$$P_{\text{right}} := \Pr[b = 1 \vee \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)], \quad (\text{Case 5})$$

$$P_{\text{right}} := \Pr[b = 1 \vee \text{Find} : b \leftarrow \mathcal{A}^{G \setminus S}(z)]. \quad (\text{Case 6})$$

Inverting a random oracle with leakage

In this section, we consider a searching problem related to a random oracle and an injective function. This problem is actually a variant of the searching problem considered in Section 4.2 of Ambainis et al. (2019), where the underlying hard problem is replaced into inverting an injective function.

Let $P : X \rightarrow Y$ be a function that satisfies the following two conditions:

- (a) For any $x \in X$, there does not exist $x' \neq x$ such that $P(x') = P(x)$.
- (b) For $x \xleftarrow{\$} X$, it is difficult to compute x from $P(x)$.

That is, P is an injective function and is hard to invert. For example, P could be the encryption algorithm of a perfect correct dPKE scheme. For a quantum algorithm $\mathcal{B}$, we define its P -inverting advantage as

$$\text{Adv}_{\text{INV-}P}(\mathcal{B}) := \Pr \left[x' = x : x' \leftarrow \mathcal{B}(P(x)), x \xleftarrow{\$} X \right].$$

Let $H : X \rightarrow Z$ be a random oracle that is independent of P . Now we consider a searching problem for an algorithm with quantum oracle access to H :

For $x \xleftarrow{\$} X$, given $H(x)$ and $P(x)$, guess the preimage x of $H(x)$.

For quantum oracle algorithm $\mathcal{A}$, we define its P -leak-inverting advantage as

$$\text{Adv}_{\text{INV-LEAK-}P}(\mathcal{A})$$

$$:= \Pr \left[x' = x : x' \leftarrow \mathcal{A}^H(H(x), P(x)), x \xleftarrow{\$} X \right].$$

Now by using our Lemma 4, we prove the following theorem, which presents a tight reduction from the hardness of inverting a random oracle with leakage to that of inverting an injective function.

Theorem 1 For any adversary $\mathcal{A}$ that makes q_H quantum queries to random oracle H , there exists an adversary $\mathcal{B}$ such that

$$\text{Adv}_{\text{INV-LEAK-}P}(\mathcal{A}) \leq 4 \cdot \text{Adv}_{\text{INV-}P}(\mathcal{B}).$$

The running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{\text{RO}} + \text{Time}(P))$.

Proof We prove this theorem by a sequence of games $\mathbf{G}_0$ to $\mathbf{G}_2$.

Game $\mathbf{G}_0$: $x \xleftarrow{\$} X$; $x' \leftarrow \mathcal{A}^H(H(x), P(x))$. This game finally outputs $\llbracket x' = x \rrbracket$.

$$\text{Adv}_{\text{INV-LEAK-}P}(\mathcal{A}) = \Pr[1 \leftarrow \mathbf{G}_0]. \quad (4)$$

Based on algorithm $\mathcal{A}$, we define a new algorithm $\mathcal{A}_1$. It first runs $\mathcal{A}$ to get the output x' , then it makes an additional query to H on x' and ignores its result, and finally outputs x' . So $\mathcal{A}_1$ makes $q_H + 1$ queries to H , and the running time of $\mathcal{A}_1$ is almost the same as $\mathcal{A}$.

Game $\mathbf{G}_1$: $x \xleftarrow{\$} X$; $x' \leftarrow \mathcal{A}_1^H(H(x), P(x))$. This game finally outputs $\llbracket x' = x \rrbracket$.

Since the additional step in $\mathcal{A}_1$ does not influence the computation of $\mathcal{A}$,

$$\Pr[1 \leftarrow \mathbf{G}_1] = \Pr[1 \leftarrow \mathbf{G}_0]. \quad (5)$$

Game $\mathbf{G}_2$: $x \xleftarrow{\$} X$; $x' \leftarrow \mathcal{A}_1^{H \setminus \{x\}}(H(x), P(x))$. Here $H \setminus \{x\}$ is an oracle that first queries $\mathcal{O}_{\{x\}}^{\text{SC}}$ and then H . This game finally outputs $\llbracket x' = x \rrbracket$.

In game $\mathbf{G}_2$, the semi-classical oracle $\mathcal{O}_{\{x\}}^{\text{SC}}$ is introduced. Let f be a function that $f(x) = x$ for any x , then by using the (case 3) of our Lemma 4 with $S := \{x\}$ and f , there exists an algorithm $\mathcal{A}_2$ with oracle access to H and $H \setminus \{x\}$ such that

$$\left| \sqrt{\Pr[1 \leftarrow \mathbf{G}_1]} - \sqrt{\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_2]} \right| \leq 2 \sqrt{\Pr \left[w = x : w \leftarrow \mathcal{A}_2^{H, H \setminus \{x\}}(H(x), P(x)), x \xleftarrow{\$} X \right]}. \quad (6)$$

Here Find is the event that the semi-classical oracle $\mathcal{O}_{\{x\}}^{SC}$ ever returns 1 in game $\mathbf{G}_2$, and the running time of $\mathcal{A}_2$ is almost the same as $\mathcal{A}_1$.

By the definition of $\mathcal{A}_1$, if its final output x' equals x , the event Find must have occurred in game $\mathbf{G}_2$. Hence, we have

$$\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_2] = 0. \quad (7)$$

Next, based on the algorithm $\mathcal{A}_2$, we construct an algorithm $\mathcal{B}$ with input $P(x)$ to compute x as follows:

1. According to the method provided in Sect. 2.2, choose a function H' to simulate a QRO with input/output space X/Z . Then choose $z \xleftarrow{\$} Z$.
2. Run the algorithm $\mathcal{A}_2$ with input $(z, P(x))$ and finally output $\mathcal{A}_2$'s output.

- (a) When $\mathcal{A}_2$ queries H with the state $\sum |x_H, y_H\rangle$, answer it by performing the unitary operation

$$\begin{aligned} & U_{H',z}|x_H, y_H\rangle \\ &= \begin{cases} |x_H, y_H \oplus H'(x_H)\rangle & \text{if } P(x_H) \neq P(x), \\ |x_H, y_H \oplus z\rangle & \text{if } P(x_H) = P(x). \end{cases} \end{aligned}$$

- (b) When $\mathcal{A}_2$ queries $H \setminus \{x\}$ with the state $\sum |x_H, y_H\rangle$, answer it by performing the following operations:

- By the definition of semi-classical oracle $\mathcal{O}_{\{x\}}^{SC}$ given in Sect. 2.2, use the function f_x to implement $\mathcal{O}_{\{x\}}^{SC}$. Here

$$f_x(x_H) = \begin{cases} 1 & \text{if } P(x_H) = P(x), \\ 0 & \text{if } P(x_H) \neq P(x). \end{cases}$$

- Perform the same operation used in answering queries to H .

Since P is an injective function, $f_x(x_H) = 1$ if and only if $x_H = x$. This means that $\mathcal{B}$ perfectly implements $\mathcal{O}_{\{x\}}^{SC}$ within time $\text{Time}(f_x) \approx O(1) \cdot \text{Time}(P)$. Hence, the running time of $\mathcal{B}$ can be bounded as

$$\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{RO} + \text{Time}(P)).$$

By the definition of $\mathcal{B}$, we have

$$\text{Adv}_{\text{INV}-P}(\mathcal{B}) = \Pr \left[w = x : w \leftarrow \mathcal{A}_2^{H, H \setminus \{x\}}(H(x), P(x)), x \xleftarrow{\$} X \right]. \quad (8)$$

Combining (4) to (8), we get

$$\text{Adv}_{\text{INV-LEAK}-P}(\mathcal{A}) \leq 4 \cdot \text{Adv}_{\text{INV}-P}(\mathcal{B}).$$

□

By using our new double-sided O2H lemma (i.e. Lemma 4), we get a tight reduction for the searching problem given above. If we had used the semi-classical O2H lemma (Ambainis et al. 2019) to prove Theorem 1, we would have gotten a non-tight reduction with a greater bound of $(q_H + 1) \cdot \text{Adv}_{\text{INV}-P}(\mathcal{B})$.

Tight proofs for worst-case to average-case transform

In this section, we prove that the OW-CPA security of transformations ACWC_0 and ACWC_1 can be tightly reduced to the OW-CPA security of the underlying PKE scheme, in the QROM.

- ACWC_0 defined in Duman et al. (2023) can transform a PKE scheme with average correctness error into a PKE scheme with worst correctness error. The drawback of ACWC_0 is that it increases the length of the ciphertext.
- ACWC_1 is a variant of ACWC defined in Duman et al. (2023). Compared with ACWC_0 , ACWC has the advantage that it avoids increasing the length of the ciphertext.

We also prove that ACWC_1 can transform a PKE scheme with average correctness error into a PKE scheme with worst correctness error, without increasing the length of the ciphertext. The definition of cryptographic primitives and some special properties used in this section are given in Appendix B.

Before we prove the security of ACWC_0 and ACWC_1 , we first give the following lemma. For a δ -average/worst-correct PKE scheme, this lemma shows that with error term 2δ , the ciphertext corresponding to a randomly chosen plaintext cannot be an encryption of another different one.

Lemma 5 *For a unique randomness recoverable and δ -average/worst-correct PKE scheme $\mathbf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ and randomness space $\mathcal{R}$, we have*

$$\Pr \left[\begin{array}{c} \exists (m' \neq m, r') \in \mathcal{M} \times \mathcal{R}, \\ \text{Enc}_{pk}(m'; r') = \text{Enc}_{pk}(m; r) \end{array} \right] \leq 2\delta,$$

where the probability is taken over $(pk, sk) \leftarrow \text{Gen}$, $m \xleftarrow{\$} \mathcal{M}$, and $r \xleftarrow{\$} \mathcal{R}$.

Gen	$\text{Enc}'_{pk}(m \in \mathcal{M})$	$\text{Dec}'_{sk}(c_1, c_2)$
1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $r_1 \xleftarrow{\$} \mathcal{M}_P, r_2 \xleftarrow{\$} \mathcal{R}_P$	1 : $r_1 := \text{Dec}_{sk}(c_1)$
2 : return (pk, sk)	2 : $c_1 := \text{Enc}_{pk}(r_1; r_2), c_2 := H(r_1) \oplus m$	2 : $m := H(r_1) \oplus c_2$
	3 : return (c_1, c_2)	3 : return m

Fig. 2 Public Key Encryption $\text{ACWC}_0[\mathbf{P}, H] = (\text{Gen}, \text{Enc}', \text{Dec}')$ with message space $\mathcal{M} := \{0, 1\}^n$. The length of the ciphertext of this scheme is n bits longer than $\mathcal{P}$

GAMES $\mathbf{G}_0\text{-}\mathbf{G}_4$	$H(x_H, y_H\rangle)$
1 : $(pk, sk) \leftarrow \text{Gen}$ // $\mathbf{G}_0\text{-}\mathbf{G}_4$	1 : return $ x_H, y_H \oplus H(x_H)\rangle$ // $\mathbf{G}_0\text{-}\mathbf{G}_2$
2 : $H \xleftarrow{\$} \Omega_H$ // $\mathbf{G}_0\text{-}\mathbf{G}_4$	2 : if $x_H = r_1^*$ // $\mathbf{G}_3$
3 : $u^* \xleftarrow{\$} \mathcal{M}$ // $\mathbf{G}_3\text{-}\mathbf{G}_4$	3 : return $ x_H, y_H \oplus u^*\rangle$
4 : $r_1^* \xleftarrow{\$} \mathcal{M}_P, r_2^* \xleftarrow{\$} \mathcal{R}_P$ // $\mathbf{G}_0\text{-}\mathbf{G}_4$	4 : else return $ x_H, y_H \oplus H(x_H)\rangle$
5 : $m^* \xleftarrow{\$} \mathcal{M}$ // $\mathbf{G}_0\text{-}\mathbf{G}_4$	5 : if $\text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c_1^*)) = c_1^*$ // $\mathbf{G}_4$
6 : $c_1^* := \text{Enc}_{pk}(r_1^*; r_2^*)$ // $\mathbf{G}_0\text{-}\mathbf{G}_4$	6 : return $ x_H, y_H \oplus u^*\rangle$
7 : $c_2^* := H(r_1^*) \oplus m^*$ // $\mathbf{G}_0\text{-}\mathbf{G}_2$	7 : else return $ x_H, y_H \oplus H(x_H)\rangle$
8 : $c_2^* := u^* \oplus m^*$ // $\mathbf{G}_3\text{-}\mathbf{G}_4$	$\mathcal{O}_{\{r_1^*\}}^{SC}(x_H, y_H\rangle)$
9 : $c^* := (c_1^*, c_2^*)$ // $\mathbf{G}_0\text{-}\mathbf{G}_4$	1 : initialize register B with $ 0\rangle$ // $\mathbf{G}_1\text{-}\mathbf{G}_4$
10 : $m' \leftarrow \mathcal{A}^H(pk, c^*)$ // $\mathbf{G}_0$	2 : perform // $\mathbf{G}_1\text{-}\mathbf{G}_3$
11 : $m' \leftarrow \mathcal{A}^{H \setminus \{r_1^*\}}(pk, c^*)$ // $\mathbf{G}_1$	3 : $ x_H, y_H\rangle 0\rangle \mapsto x_H, y_H\rangle f_{r_1^*}(x_H)\rangle$
12 : $r_1' \leftarrow \mathcal{A}_1^{H, H \setminus \{r_1^*\}}(pk, c^*)$ // $\mathbf{G}_2\text{-}\mathbf{G}_4$	4 : perform // $\mathbf{G}_4$
13 : return $\llbracket m' = m^* \rrbracket$ // $\mathbf{G}_0\text{-}\mathbf{G}_1$	5 : $ x_H, y_H\rangle 0\rangle \mapsto x_H, y_H\rangle f_{r_1'}'(x_H)\rangle$
14 : return $\llbracket r_1' = r_1^* \rrbracket$ // $\mathbf{G}_2\text{-}\mathbf{G}_4$	6 : measure register B , return the result // $\mathbf{G}_1\text{-}\mathbf{G}_4$

Fig. 3 Games $\mathbf{G}_0$ to $\mathbf{G}_4$ in the proof of Theorem 2. The oracles H and $\mathcal{O}_{\{r_1^*\}}^{SC}$ can both be queried in the superposition, while in this figure we just write the input state of H and $\mathcal{O}_{\{r_1^*\}}^{SC}$ as $|x_H, y_H\rangle$ for brevity

Proof The proof of this lemma is given in Appendix C.1. $\square$

Simple transformation ACWC_0 with redundancy

Let $\mathbf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a PKE scheme with message space $\mathcal{M}_P$ and randomness space $\mathcal{R}_P$. Let $H : \mathcal{M}_P \rightarrow \{0, 1\}^n$ be a hash function. We associate $\text{ACWC}_0[\mathbf{P}, H] := (\text{Gen}, \text{Enc}', \text{Dec}')$. The constituting algorithms of $\text{ACWC}_0[\mathbf{P}, H]$ are given in Fig. 2.

Theorem 2 (OW-CPA of $\text{ACWC}_0[\mathbf{P}, H]$) *Suppose $\mathbf{P}$ is unique randomness recoverable and δ -average-correct. Let $\mathcal{A}$ be a OW-CPA adversary against $\text{ACWC}_0[\mathbf{P}, H]$,*

making q_H quantum queries to random oracle H . Then there exists a OW-CPA adversary $\mathcal{B}$ against $\mathbf{P}$ such that

$$\text{Adv}_{\text{ACWC}_0[\mathbf{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) \leq 8 \cdot \text{Adv}_{\mathbf{P}}^{\text{OW-CPA}}(\mathcal{B}) + 16 \cdot \delta + \frac{2}{|\mathcal{M}|}.$$

Let Rec be the recover algorithm of $\mathbf{P}$, the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$.

Proof We prove this theorem by a sequence of games (see also Fig. 3).

Game $\mathbf{G}_0$: This is the OW-CPA game of $\text{ACWC}_0[\mathbf{P}, H]$ with adversary $\mathcal{A}$.

$$\text{Adv}_{\text{ACWC}_0[\mathcal{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) = \Pr[1 \leftarrow \mathbf{G}_0]. \quad (9)$$

Game $\mathbf{G}_1$: This game is the same as game $\mathbf{G}_0$, except that the oracle H is changed into $H \setminus \{r_1^*\}$. Here $H \setminus \{r_1^*\}$ is an oracle that first queries the semi-classical oracle $\mathcal{O}_{\{r_1^*\}}^{\text{SC}}$ and then H . $\mathcal{O}_{\{r_1^*\}}^{\text{SC}}$ in this game is simulated by using the function

$$f_{r_1^*}(x_H) := \begin{cases} 1 & \text{if } x_H = r_1^*, \\ 0 & \text{if } x_H \neq r_1^*. \end{cases} \quad (10)$$

According to Lemma 4, where $S := \{r_1^*\}$ and $f : \mathcal{M}_{\mathcal{P}} \rightarrow \mathcal{M}_{\mathcal{P}}$ is an identical function, there exists an adversary $\mathcal{A}_1$ with quantum oracle access to H and $H \setminus \{r_1^*\}$ such that

$$\left| \sqrt{\Pr[1 \leftarrow \mathbf{G}_0]} - \sqrt{\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_1]} \right| \leq 2\sqrt{\Pr[r_1' = r_1^* : r_1' \leftarrow \mathcal{A}_1^{H, H \setminus \{r_1^*\}}(pk, c^*)]}. \quad (11)$$

Here (pk, c^*) is generated as the same way in game $\mathbf{G}_1$, Find is the event that the semi-classical oracle $\mathcal{O}_{\{r_1^*\}}^{\text{SC}}$ ever returns 1 in game $\mathbf{G}_1$, and the running time of $\mathcal{A}_1$ is almost the same as $\mathcal{A}$.

If Find does not occur in game $\mathbf{G}_1$, $\mathcal{A}$ will only query H with the state $\sum_{x_H \neq r_1^*, y_H} |x_H, y_H\rangle$, then both $H(r_1^*)$ and $H(r_1^*) \oplus m^*$ are uniformly random in $\mathcal{A}$'s view. That is, if Find does not occur in game $\mathbf{G}_1$, $\mathcal{A}$'s input (pk, c^*) is independent with the challenge plaintext m^* . Thus, we have

$$\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_1] \leq \frac{1}{|\mathcal{M}|}. \quad (12)$$

Combining (11), (12), and the fact that $(a + b)^2 \leq 2a^2 + 2b^2$, we obtain

$$\Pr[1 \leftarrow \mathbf{G}_0] \leq \frac{2}{|\mathcal{M}|} + 8 \cdot \Pr[r_1' = r_1^* : r_1' \leftarrow \mathcal{A}_1^{H, H \setminus \{r_1^*\}}(pk, c^*)]. \quad (13)$$

Game $\mathbf{G}_2$: This game is the same as game $\mathbf{G}_1$, except that the adversary $\mathcal{A}$ is changed into the adversary $\mathcal{A}_1$.

$$\Pr[1 \leftarrow \mathbf{G}_2] = \Pr[r_1' = r_1^* : r_1' \leftarrow \mathcal{A}_1^{H, H \setminus \{r_1^*\}}(pk, c^*)]. \quad (14)$$

Game $\mathbf{G}_3$: Different with game $\mathbf{G}_2$, $H(r_1^*)$ in game $\mathbf{G}_3$ is reprogrammed into u^* , where $u^* \xleftarrow{\$} \mathcal{M}$ is sampled at the beginning of the game.

Obviously, the change in game $\mathbf{G}_3$ is only conceptual, then

$$\Pr[1 \leftarrow \mathbf{G}_2] = \Pr[1 \leftarrow \mathbf{G}_3]. \quad (15)$$

Game $\mathbf{G}_4$: This game is the same as game $\mathbf{G}_3$, except that the simulation of oracle H and $\mathcal{O}_{\{r_1^*\}}^{\text{SC}}$ is changed.

1. Simulation of H : If the input x_H satisfies $\text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c_1^*)) = c_1^*$, u^* will be returned. Otherwise, $H(x_H)$ is returned.
2. Simulation of $\mathcal{O}_{\{r_1^*\}}^{\text{SC}}$: The function $f_{r_1^*}$ (as shown in (10)) is changed into

$$f_{r_1^*}'(x_H) = \begin{cases} 1 & \text{if } \text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c_1^*)) = c_1^*, \\ 0 & \text{if } \text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c_1^*)) \neq c_1^*. \end{cases} \quad (16)$$

Let Bad be the event that there exists $r_1 \neq r_1^*$ and $r_2 \in \mathcal{R}_{\mathcal{P}}$ satisfying $\text{Enc}_{pk}(r_1; r_2) = c_1^*$. We can conclude that, if Bad does not occur in game $\mathbf{G}_4$, $\text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c_1^*))$ equals to c_1^* iff x_H equals to r_1^* . This means that game $\mathbf{G}_3$ and game $\mathbf{G}_4$ are the same if Bad does not occur. By using the difference lemma in Shoup (2004),

$$|\Pr[1 \leftarrow \mathbf{G}_3] - \Pr[1 \leftarrow \mathbf{G}_4]| \leq \Pr[\text{Bad}]. \quad (17)$$

Since $\mathcal{P}$ is δ -average-correct, by using Lemma 5, we have

$$\Pr[\text{Bad}] \leq 2 \cdot \delta. \quad (18)$$

Now we define a OW-CPA adversary $\mathcal{B}$ against $\mathcal{P}$ as follows:

1. The input is $(pk, c_1^* := \text{Enc}_{pk}(r_1^*; r_2^*))$, where $r_1^* \xleftarrow{\$} \mathcal{M}_{\mathcal{P}}$ and $r_2^* \xleftarrow{\$} \mathcal{R}_{\mathcal{P}}$.
2. According to the method provided in Sect. 2.2, we choose a function H to simulate a QRO with input/output space $\mathcal{M}_{\mathcal{P}}/\mathcal{M}$.
3. Choose $m^* \xleftarrow{\$} \mathcal{M}$ and $u^* \xleftarrow{\$} \mathcal{M}$. Let $c_2^* = u^* \oplus m^*$.
4. Run the adversary $\mathcal{A}_1^{H, H \setminus \{r_1^*\}}(pk, (c_1^*, c_2^*))$ in game $\mathbf{G}_4$. When $\mathcal{A}_1$ queries H or $H \setminus \{r_1^*\}$, answer it in the same way as in game $\mathbf{G}_4$.

Note that the function $f_{r_1^*}'$ (as shown in (16)) can be computed by using the algorithms Enc and Rec, the running time of adversary $\mathcal{B}$ can be bounded as $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$. Obviously,

Gen	$\text{Enc}'_{pk}(m \in \mathcal{M})$	$\text{Dec}'_{sk}(c)$
1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $M_1 \xleftarrow{\$} \mathcal{M}_1, r \xleftarrow{\$} \mathcal{R}$	1 : $(M_1, M_2) := \text{Dec}_{sk}(c)$
2 : return (pk, sk)	2 : $M_2 := m \oplus H(M_1)$	2 : $m := M_2 \oplus H(M_1)$
	3 : $c := \text{Enc}_{pk}(M_1 M_2; r)$	3 : return m
	4 : return c	

Fig. 4 Public Key Encryption $\text{ACWC}[\mathbf{P}, H] = (\text{Gen}, \text{Enc}', \text{Dec}')$ with message space $\mathcal{M} := \mathcal{M}_2$. This scheme has the same ciphertext length as the scheme $\mathbf{P}$

Gen	$\text{Enc}'_{pk}(m_1 m_2 \in \mathcal{M} \times \mathcal{R}_2)$	$\text{Dec}'_{sk}(c)$
1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $r_1 \xleftarrow{\$} \mathcal{R}_1$	1 : $m_1 \leftarrow \text{Dec}_{sk}(c)$
2 : return (pk, sk)	2 : $c := \text{Enc}_{pk}(m_1; r_1 G(m_1) \oplus m_2)$	2 : $r_1 r_2 \leftarrow \text{Rec}_{pk}(m_1, c)$
	3 : return c	3 : $m_2 := r_2 \oplus G(m_1)$
		4 : return $m_1 m_2$

Fig. 5 Public Key Encryption $\text{POTO}[\mathbf{P}, G] = (\text{Gen}, \text{Enc}', \text{Dec}')$ with message space $\mathcal{M} \times \mathcal{R}_2$. This scheme has the same ciphertext length as the scheme. Here Rec is the recover algorithm of (unique randomness recoverable) scheme $\mathbf{P}$

$$\text{Adv}_{\mathbf{P}}^{\text{OW-CPA}}(\mathcal{B}) = \Pr[1 \leftarrow \mathbf{G}_4]. \quad (19)$$

Combining (9), (13) to (15) and (17) to (19), we get

$$\text{Adv}_{\text{ACWC}_0[\mathbf{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) \leq 8 \cdot \text{Adv}_{\mathbf{P}}^{\text{OW-CPA}}(\mathcal{B}) + 16 \cdot \delta + \frac{2}{|\mathcal{M}|}.$$

□

Transformation ACWC_1 without redundancy

In this section, we consider the transformation $\text{ACWC}_1 := \text{ACWC} \circ \text{POTO}$, it is a variant of ACWC defined in Duman et al. (2023). In the QROM, we give the following two proofs:

1. The OW-CPA security of ACWC can be tightly reduced to the POW-CPA security of the underlying PKE scheme.
2. The POW-CPA security of POTO can be tightly reduced to the OW-CPA security of the underlying PKE scheme.

Combining the above two proofs, we prove that the OW-CPA security of ACWC_1 can be tightly reduced to the OW-CPA security of the underlying PKE scheme.

Transformation ACWC : Let $\mathbf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a PKE scheme with message space $\mathcal{M}_1 \times \mathcal{M}_2$

($\mathcal{M}_2 := \{0, 1\}^n$) and randomness space $\mathcal{R}$. Let $H : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a hash function. We associate $\text{ACWC}[\mathbf{P}, H] := (\text{Gen}, \text{Enc}', \text{Dec}')$. The constituting algorithms of $\text{ACWC}[\mathbf{P}, H]$ are given in Fig. 4.

Theorem 3 (OW-CPA of $\text{ACWC}[\mathbf{P}, H]$)

Suppose $\mathbf{P}$ is randomness recoverable and δ -average-correct. Let $\mathcal{A}$ be an OW-CPA adversary against $\text{ACWC}[\mathbf{P}, H]$, making q_H quantum queries to random oracle H . Then there exists a POW-CPA adversary $\mathcal{B}$ against $\mathbf{P}$ such that

$$\text{Adv}_{\text{ACWC}[\mathbf{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) \leq 8 \cdot \text{Adv}_{\mathbf{P}}^{\text{POW-CPA}}(\mathcal{B}) + 16 \cdot \delta + \frac{2}{|\mathcal{M}|}.$$

Let Rec be the recover algorithm of $\mathbf{P}$, the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$.

Proof The proof of this theorem is similar to the proof of Theorem 2, and we give it in Appendix C.2. □

Transformation POTO : Let $\mathbf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a unique randomness recoverable PKE scheme with message space $\mathcal{M}$ and randomness space $\mathcal{R} := \mathcal{R}_1 \times \mathcal{R}_2$ ($\mathcal{R}_2 := \{0, 1\}^n$). Let $G : \mathcal{M} \rightarrow \mathcal{R}_2$ be a hash function.

GAMES $\mathbf{G_0-G_5}$		$G(x_G, y_G\rangle)$ (Let $r_1 r_2 = \text{Rec}_{pk}(x_G, c^*)$)	
1 : $(pk, sk) \leftarrow \text{Gen}$	// $\mathbf{G_0-G_5}$	1 : return $ x_G, y_G \oplus G(x_G)\rangle$	// $\mathbf{G_0-G_3}$
2 : $G \xleftarrow{\$} \Omega_G$	// $\mathbf{G_0-G_5}$	2 : if $x_G = m_1^*$	// $\mathbf{G_4}$
3 : $m_1^* \xleftarrow{\$} \mathcal{M}$	// $\mathbf{G_0-G_5}$	3 : return $ x_G, y_G \oplus (r_2^* \oplus m_2^*)\rangle$	
4 : $m_2^* \xleftarrow{\$} \mathcal{R}_2$	// $\mathbf{G_0-G_5}$	4 : else return $ x_G, y_G \oplus G(x_G)\rangle$	
5 : $r_1^* \xleftarrow{\$} \mathcal{R}_1$	// $\mathbf{G_0-G_5}$	5 : if $\text{Enc}_{pk}(x_G; \text{Rec}_{pk}(x_G, c^*)) = c^*$	// $\mathbf{G_5}$
6 : $r_2^* \xleftarrow{\$} \mathcal{R}_2$	// $\mathbf{G_4-G_5}$	6 : return $ x_G, y_G \oplus (r_2 \oplus m_2^*)\rangle$	
7 : $c^* := \text{Enc}_{pk}(m_1^*; r_1^* G(m_1^*) \oplus m_2^*)$	// $\mathbf{G_0-G_3}$	7 : else return $ x_G, y_G \oplus G(x_G)\rangle$	
8 : $c^* := \text{Enc}_{pk}(m_1^*; r_1^* r_2^*)$	// $\mathbf{G_4-G_5}$	$\mathcal{O}_{\{m_1^*\}}^{SC}(x_G, y_G\rangle)$	
9 : $m_1' \leftarrow \mathcal{A}_1^G(pk, c^*, m_2^*)$	// $\mathbf{G_0}$	1 : initialize register B with $ 0\rangle$	// $\mathbf{G_2-G_5}$
10 : $m_1' \leftarrow \mathcal{A}_1^G(pk, c^*, m_2^*)$	// $\mathbf{G_1}$	2 : perform	// $\mathbf{G_2-G_4}$
11 : $m_1' \leftarrow \mathcal{A}_1^{G \setminus \{m_1^*\}}(pk, c^*, m_2^*)$	// $\mathbf{G_2}$	3 : $ x_G, y_G\rangle 0\rangle \mapsto x_G, y_G\rangle f_{m_1^*}(x_G)\rangle$	
12 : $m_1' \leftarrow \mathcal{A}_2^{G, G \setminus \{m_1^*\}}(pk, c^*, m_2^*)$	// $\mathbf{G_3-G_5}$	4 : perform	// $\mathbf{G_5}$
13 : return $\llbracket m_1' = m_1^* \rrbracket$	// $\mathbf{G_0-G_5}$	5 : $ x_G, y_G\rangle 0\rangle \mapsto x_G, y_G\rangle f_{m_1^*}'(x_G)\rangle$	
		6 : measure register B , return the result	// $\mathbf{G_2-G_5}$

Fig. 6 Games $\mathbf{G_0}$ to $\mathbf{G_5}$ in the proof of Theorem 4. The oracles G and $\mathcal{O}_{\{m_1^*\}}^{SC}$ can both be queried in the superposition, while in this figure we just write the input state of G and $\mathcal{O}_{\{m_1^*\}}^{SC}$ as $|x_G, y_G\rangle$ for brevity

We associate $\text{POTO}[\mathbf{P}, G] := (\text{Gen}, \text{Enc}', \text{Dec}')$. The constituting algorithms of $\text{POTO}[\mathbf{P}, G]$ are given in Fig. 5.

Before we prove the security of POTO , we first prove the following lemma, which shows that POTO remains some essential properties of the underlying PKE scheme $\mathbf{P}$. The proof is given in Appendix C.3.

Lemma 6 *For unique randomness recoverable PKE scheme $\mathbf{P}$, if it is also δ -average-correct, then $\text{POTO}[\mathbf{P}, G]$ is unique randomness recoverable and δ -average-correct.*

Theorem 4 [POW-CPA of $\text{POTO}[\mathbf{P}, G]$] *Suppose $\mathbf{P}$ is unique randomness recoverable and δ -average-correct. Let $\mathcal{A}$ be a POW-CPA adversary against $\text{POTO}[\mathbf{P}, G]$, making q_G quantum queries to random oracle G . Then there exists a OW-CPA adversary $\mathcal{B}$ against $\mathbf{P}$ such that*

$$\text{Adv}_{\text{POTO}[\mathbf{P}, G]}^{\text{POW-CPA}}(\mathcal{A}) \leq 4 \cdot \text{Adv}_{\mathbf{P}}^{\text{OW-CPA}}(\mathcal{B}) + 8 \cdot \delta.$$

Let Rec be the recover algorithm of $\mathbf{P}$, the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_G) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$.

Proof To prove this theorem, we define a sequence of games (see also Fig. 6).

Game $\mathbf{G_0}$: This is the POW-CPA game of $\text{POTO}[\mathbf{P}, G]$ with adversary $\mathcal{A}$.

$$\text{Adv}_{\text{POTO}[\mathbf{P}, G]}^{\text{POW-CPA}}(\mathcal{A}) = \Pr[1 \leftarrow \mathbf{G_0}]. \quad (20)$$

Game $\mathbf{G_1}$: In this game, we change $\mathcal{A}$ into $\mathcal{A}_1$, which first runs $\mathcal{A}$ to get the outcome m_1' , then queries G on m_1' and ignores its result. Finally it returns m_1' as its output.

By the definition of $\mathcal{A}_1$, it makes $q_G + 1$ queries to G , and the running time of $\mathcal{A}_1$ is almost the same as $\mathcal{A}$. Note that the additional step of $\mathcal{A}_1$ does not influence the computation of $\mathcal{A}$, we have

$$\Pr[1 \leftarrow \mathbf{G_0}] = \Pr[1 \leftarrow \mathbf{G_1}]. \quad (21)$$

Game $\mathbf{G_2}$: Compared with game $\mathbf{G_1}$, the oracle G is changed into $G \setminus \{m_1^*\}$. Here $G \setminus \{m_1^*\}$ is an oracle that first queries the semi-classical oracle $\mathcal{O}_{\{m_1^*\}}^{SC}$ and then G . $\mathcal{O}_{\{m_1^*\}}^{SC}$ in this game is simulated by using the function

$$f_{m_1^*}(x_G) := \begin{cases} 1 & \text{if } x_G = m_1^*, \\ 0 & \text{if } x_G \neq m_1^*. \end{cases} \quad (22)$$

According to Lemma 4, where $S := \{m_1^*\}$ and $f : \mathcal{M} \rightarrow \mathcal{M}$ is an identical function, there exists an adversary $\mathcal{A}_2$ with quantum oracle access to G and $G \setminus \{m_1^*\}$ such that

$$\begin{aligned} & \left| \sqrt{\Pr[1 \leftarrow \mathbf{G_1}]} - \sqrt{\Pr[b \leftarrow \mathbf{G_2} \wedge \neg \text{Find} : b \leftarrow \mathbf{G_2}]} \right| \\ & \leq 2 \sqrt{\Pr[m_1^* = m_1' : m_1' \leftarrow \mathcal{A}_1^{G, G \setminus \{m_1^*\}}(pk, c^*, m_2^*)]}. \end{aligned} \quad (23)$$

Here (pk, c^*, m_2^*) is generated as the same way in game $\mathbf{G}_2$, Find is the event that the semi-classical oracle $\mathcal{O}_{\{m_1^*\}}^{SC}$ ever returns 1 in game $\mathbf{G}_2$, and the running time of $\mathcal{A}_2$ is almost the same as $\mathcal{A}_1$.

In game $\mathbf{G}_2$, if $\mathcal{A}_1$ returns m_1^* , the event Find must have occurred by the definition of $\mathcal{A}_1$. Hence we have

$$\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_2] = 0. \quad (24)$$

Game $\mathbf{G}_3$: This game is the same as game $\mathbf{G}_2$, except that the adversary $\mathcal{A}_1$ is changed into $\mathcal{A}_2$, which has oracle access to G and $G \setminus \{m_1^*\}$.

$$\begin{aligned} \Pr[m_1' = m_1^* : m_1' \leftarrow \mathcal{A}_2^{G, G \setminus \{m_1^*\}}(pk, c^*, m_2^*)] \\ = \Pr[1 \leftarrow \mathbf{G}_3]. \end{aligned} \quad (25)$$

Game $\mathbf{G}_4$: In this game, $G(m_1^*)$ is reprogrammed into $r_2^* \oplus m_2^*$, where $r_2^* \xleftarrow{\$} \mathcal{R}_2$ is chosen at the beginning of the game.

Since $r_2^* = G(m_1^*) \oplus m_2^*$ still holds and $G(m_1^*)$ is uniformly random in game $\mathbf{G}_4$, the change from game $\mathbf{G}_3$ to $\mathbf{G}_4$ is only conceptual, then

$$\Pr[1 \leftarrow \mathbf{G}_3] = \Pr[1 \leftarrow \mathbf{G}_4]. \quad (26)$$

Game $\mathbf{G}_5$: This game is the same as game $\mathbf{G}_4$, except that the simulation of oracle G and $\mathcal{O}_{\{m_1^*\}}^{SC}$ is changed.

1. Simulation of G : For the input x_G , let $r_1 || r_2 = \text{Rec}_{pk}(x_G, c^*)$. If the input x_G satisfies $\text{Enc}_{pk}(x_G; r_1 || r_2) = c^*$, $r_2 \oplus m_2^*$ will be returned. Otherwise, $G(x_G)$ is returned.
2. Simulation of $\mathcal{O}_{\{m_1^*\}}^{SC}$: The function $f_{m_1^*}$ (as shown in (22)) is changed into

$$f_{m_1^*}'(x_G) = \begin{cases} 1 & \text{if } \text{Enc}_{pk}(x_G; \text{Rec}_{pk}(x_G, c^*)) = c^* \\ 0 & \text{if } \text{Enc}_{pk}(x_G; \text{Rec}_{pk}(x_G, c^*)) \neq c^*. \end{cases} \quad (27)$$

Let Bad be the event that there exists $m \neq m_1^*$ and $r \in \mathcal{R}$ satisfying $\text{Enc}_{pk}(m; r) = c^*$. Consider the situation when Bad does not occur in game $\mathbf{G}_5$, then $\text{Enc}_{pk}(x_G; \text{Rec}_{pk}(x_G, c^*)) = c^*$ holds iff x_G equals to m_1^* . Also, due to the unique randomness recoverability of $\mathbb{P}$, we have $\text{Rec}_{pk}(m_1^*, c^*) = r_2^*$. These indicate that game $\mathbf{G}_5$ is exactly the same as game $\mathbf{G}_4$ if event Bad does not occur. Hence, by using the difference lemma in Shoup (2004), we have

$$|\Pr[1 \leftarrow \mathbf{G}_4] - \Pr[1 \leftarrow \mathbf{G}_5]| \leq \Pr[\text{Bad}]. \quad (28)$$

Since $\mathbb{P}$ is δ -average-correct, by using Lemma 5, we have

$$\Pr[\text{Bad}] \leq 2 \cdot \delta. \quad (29)$$

Now we define a OW-CPA adversary $\mathcal{B}$ against $\mathbb{P}$ as follows:

1. The input is $(pk, c^* := \text{Enc}_{pk}(m_1^*, r^*))$, where $m_1^* \xleftarrow{\$} \mathcal{M}$ and $r^* \xleftarrow{\$} \mathcal{R}$.
2. According to the method provided in Sect. 2.2, we choose a function G to simulate a QRO with input/output space $\mathcal{M}/\mathcal{R}_2$. Then choose $m_2^* \xleftarrow{\$} \mathcal{R}_2$.
3. Run the adversary $\mathcal{A}_2^{G, G \setminus \{m_1^*\}}(pk, c^*, m_2^*)$ in game $\mathbf{G}_5$. When $\mathcal{A}_2$ queries G or $G \setminus \{m_1^*\}$, answer it in the same way as in game $\mathbf{G}_5$.

Note that the function $f_{m_1^*}'$ (as shown in (27)) can be computed by using the algorithms Enc and Rec, the running time of $\mathcal{B}$ can be bounded as $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_G) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$. Obviously,

$$\text{Adv}_{\mathbb{P}}^{\text{OW-CPA}}(\mathcal{B}) = \Pr[1 \leftarrow \mathbf{G}_5]. \quad (30)$$

Combining (20), (21), (23) to (26) and (28) to (30), we get

$$\text{Adv}_{\text{POTO}[\mathbb{P}, G]}^{\text{POW-CPA}}(\mathcal{A}) \leq 4 \cdot \text{Adv}_{\mathbb{P}}^{\text{OW-CPA}}(\mathcal{B}) + 8 \cdot \delta. \quad \square$$

By the fact that unique randomness recoverable PKE scheme is also randomness recoverable, we combine Theorem 3, Lemma 6 and Theorem 4 to obtain the following theorem, which shows that the OW-CPA security of our new transformation $\text{ACWC}_1 := \text{ACWC} \circ \text{POTO}$ can be tightly reduced to the OW-CPA security of the underlying PKE scheme.

Theorem 5 [OW-CPA of $\text{ACWC}_1[\mathbb{P}, G, H]$] *Let $\mathbb{P}$ be a PKE scheme with message space $\mathcal{M}_1 \times \mathcal{M}_2$, suppose $\mathbb{P}$ is unique randomness recoverable and δ -average-correct. Let $\mathcal{A}$ be a OW-CPA adversary against $\text{ACWC}_1[\mathbb{P}, G, H]$, making q_G and q_H quantum queries to random oracle G and H , respectively. Then there exists a OW-CPA adversary $\mathcal{B}$ against $\mathbb{P}$ such that*

$$\text{Adv}_{\text{ACWC}_1[\mathbb{P}, G, H]}^{\text{OW-CPA}}(\mathcal{A}) \leq 32 \cdot \text{Adv}_{\mathbb{P}}^{\text{OW-CPA}}(\mathcal{B}) + 80 \cdot \delta + \frac{2}{|\mathcal{M}_2|}.$$

Let Rec be the recover algorithm of $\mathbb{P}$, the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_G + q_H) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$.

$\text{Gen}^\perp$	$\text{Enca}_m(pk)$	$\text{Deca}_m^\perp(sk, c)$	$\text{Deca}_m^\perp(sk', c)$
1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $m \xleftarrow{\$} \mathcal{M}$	1 : $m' := \text{Dec}_{sk}(c)$	
2 : $s \xleftarrow{\$} \mathcal{K}^{prf}$	2 : $c := \text{Enc}_{pk}(m; H(m))$	2 : if $m' = \perp$ or $c \neq \text{Enc}_{pk}(m'; H(m'))$	
3 : $sk' := (sk, s)$	3 : $K = G(m)$	3 : return $K := \perp$; $K := f_{prf}(s, c)$	
4 : return (pk, sk')	4 : return (K, c)	4 : else return $K := G(m')$	

Fig. 7 Key Encapsulation Mechanism $\text{KEM}_m^\perp$ and $\text{KEM}_m^\perp$

The Lemma 3.6 in Duman et al. (2023) shows that, ACWC can transform a PKE scheme with average correctness error into a PKE scheme with worst correctness error. Hence, by using Lemma 6 given above, our new transformation $\text{ACWC}_1 := \text{ACWC} \circ \text{POTO}$ can also transform a PKE scheme with average correctness error into a PKE scheme with worst correctness error.

Proofs of $\text{FO}_m^\perp$ and $\text{FO}_m^\perp$

In this section, we consider the FO-like transformations $\text{FO}_m^\perp$ ¹³ and $\text{FO}_m^\perp$ defined in Hofheinz et al. (2017) that are used to construct a KEM scheme from a PKE scheme, we prove their IND-CCA security in the ROM and QROM. The formal definition of cryptographic primitives and some special properties used in this section are given in Appendix B.

Let $\mathsf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a PKE scheme with message space $\mathcal{M}$ and randomness space $\mathcal{R}$. Let $H : \mathcal{M} \rightarrow \mathcal{R}$ and $G : \mathcal{M} \rightarrow \mathcal{K}$ be hash functions, where $\mathcal{K}$ denotes the key space. Let f_{prf} be a PRF with key space $\mathcal{K}^{prf}$. We associate

$$\text{KEM}_m^\perp := \text{FO}_m^\perp[\mathsf{P}, H, G] = (\text{Gen}, \text{Enca}_m, \text{Deca}_m^\perp)$$

and

$$\text{KEM}_m^\perp := \text{FO}_m^\perp[\mathsf{P}, H, G, f_{prf}] = (\text{Gen}^\perp, \text{Enca}_m, \text{Deca}_m^\perp).$$

The constituting algorithms of $\text{KEM}_m^\perp$ and $\text{KEM}_m^\perp$ are given in Fig. 7.

Moreover, as defined in Hofheinz et al. (2017), we have $\text{FO}_m^\perp[\mathsf{P}, H, G] = \text{U}_m^\perp[\text{T}[\mathsf{P}, H], G]$ and $\text{FO}_m^\perp[\mathsf{P}, H, G, f_{prf}] = \text{U}_m^\perp[\text{T}[\mathsf{P}, H], G, f_{prf}]$. Here T , $\text{U}_m^\perp$ and $\text{U}_m^\perp$ are called the modular FO transformations.

Before we prove the IND-CCA security of $\text{FO}_m^\perp$ and $\text{FO}_m^\perp$, we first prove the following lemma. This lemma shows that, if the underlying PKE scheme is unique randomness recoverable, then the OW-CPA security of T can be tightly reduced to the OW-CPA security of the underlying PKE scheme, in the QROM. Its proof is similar to Theorem 4, and is given in Appendix C.4.

Lemma 7 [OW-CPA of $\text{T}[\mathsf{P}, H]$] Suppose P is unique randomness recoverable and δ -average/worst-correct. Let $\mathcal{A}$ be a OW-CPA adversary against $\text{T}[\mathsf{P}, H]$, making q_H quantum queries to random oracle H . Then there exists a OW-CPA adversary $\mathcal{B}$ against P such that

$$\text{Adv}_{\text{T}[\mathsf{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) \leq 4 \cdot \text{Adv}_{\mathsf{P}}^{\text{OW-CPA}}(\mathcal{B}) + 8 \cdot \delta.$$

Let Rec be the recover algorithm of P , the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$.

Proofs of $\text{KEM}_m^\perp$ and $\text{KEM}_m^\perp$

We first introduce the following lemma, which shows that the IND-CCA security of $\text{KEM}_m^\perp$ can be reduced to its IND-CPA security, in the QROM.

Lemma 8 (IND-CCA of $\text{KEM}_m^\perp$; (Ge et al. 2023) Theorem 2) Suppose P is δ -worst-correct and weakly γ -spread. Let $\mathcal{A}$ be an IND-CCA adversary against $\text{KEM}_m^\perp$, making q_D , q_H and q_G queries to the decapsulation oracle, the random oracle H and the random oracle G , respectively. Let d_H (resp. d_G) be the query depth of $\mathcal{A}$'s H (resp. G) queries. Then there exists an IND-CPA adversary $\mathcal{B}$ (in the QROM) against $\text{KEM}_m^\perp$ such that

$$\begin{aligned} \text{Adv}_{\text{KEM}_m^\perp}^{\text{IND-CCA}}(\mathcal{A}) \\ \leq \text{Adv}_{\text{KEM}_m^\perp}^{\text{IND-CPA}}(\mathcal{B}) + O(q_H) \cdot \sqrt{\delta} + O(q_D) \cdot 2^{-\gamma/2}. \end{aligned}$$

The adversary $\mathcal{B}$ makes $2q_H$ (resp. $q_G + q_D$) quantum queries to H (resp. G). The query depth of $\mathcal{A}_1$ to H (resp. G) is $2d_H$ (resp. $d_G + q_D$). The running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H q_D + q_H^2)$.

Next we prove that, the IND-CPA security of $\text{KEM}_m^\perp$ can be reduced to the OW-CPA security of P , in the QROM.

Theorem 6 (IND-CPA of $\text{KEM}_m^\perp$) Suppose P is δ -worst-correct and unique randomness recoverable. Let $\mathcal{A}$ be an IND-CPA adversary against $\text{KEM}_m^\perp$, making q_H and q_G

¹³ Here $\text{FO}_m^\perp$ is actually a generic version defined in Jiang et al. (2018).

queries to random oracles H and G , respectively. Let d_H (resp. d_G) be the query depth of $\mathcal{A}$'s H (resp. G) queries. Then there exists a OW-CPA adversary $\mathcal{B}$ against $\mathcal{P}$ such that

$$\text{Adv}_{\text{KEM}_m^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq 8d_G \cdot \text{Adv}_{\mathcal{P}}^{\text{OW-CPA}}(\mathcal{B}) + 36d_G \cdot \delta.$$

Let Rec be the recover algorithm of $\mathcal{P}$, the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx 2 \cdot \text{Time}(\mathcal{A}) + O(q_H + q_G) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$.

Proof We first introduce the following lemma, which shows that the IND-CPA security of $\text{U}_m^\perp$ can be reduced to the OW-CPA security of the underlying DPKE schemes, in the QROM.

Lemma 9 [IND-CPA of $\text{U}_m^\perp[\text{DPKE}, G]$; (Ge et al. 2023) Theorem 3] Let DPKE be a dPKE scheme with message space $\mathcal{M}$. Let $\mathcal{A}$ be an IND-CPA adversary against $\text{U}_m^\perp[\text{DPKE}, G]$, making q_G queries to the random oracle G with query depth d_G . Then there exists a OW-CPA adversary $\mathcal{A}_1$ against DPKE such that

$$\begin{aligned} \text{Adv}_{\text{U}_m^\perp[\text{DPKE}, G]}^{\text{IND-CPA}}(\mathcal{A}) \\ \leq 2d_G \cdot \text{Adv}_{\text{DPKE}}^{\text{OW-CPA}}(\mathcal{A}_1) + 2d_G \cdot \Pr[E_{\text{DPKE}}]. \end{aligned}$$

Here E_{DPKE} is the event that $\exists m' \neq m$ such that $\text{dEnc}_{pk}(m) = \text{dEnc}_{pk}(m')$, where $m \xleftarrow{\$} \mathcal{M}$ and dEnc is the encryption algorithm of DPKE. The running time of $\mathcal{A}_1$ is $\text{Time}(\mathcal{A}_1) \approx 2 \cdot \text{Time}(\mathcal{A}) + O(q_G) \cdot t_{\text{RO}}$.

Now, we have

$$\begin{aligned} \text{Adv}_{\text{KEM}_m^\perp}^{\text{IND-CPA}}(\mathcal{A}) &= \text{Adv}_{\text{U}_m^\perp[\text{T}[P, H], G]}^{\text{IND-CPA}}(\mathcal{A}) \\ &\stackrel{(a)}{\leq} 2d_G \cdot \text{Adv}_{\text{T}[P, H]}^{\text{OW-CPA}}(\mathcal{A}_1) + 2d_G \cdot \Pr[E_{\text{T}[P, H]}] \\ &\stackrel{(b)}{\leq} 2d_G \cdot \text{Adv}_{\text{T}[P, H]}^{\text{OW-CPA}}(\mathcal{A}_1) + 4d_G \cdot \delta \\ &\stackrel{(c)}{\leq} 8d_G \cdot \text{Adv}_{\mathcal{P}}^{\text{OW-CPA}}(\mathcal{B}) + 20d_G \cdot \delta. \end{aligned}$$

Here (a) and (c) use Lemma 9 and Lemma 7, respectively. (b) uses Lemma 5, which implies that $\Pr[E_{\text{T}[P, H]}]$ has an upper bound $2 \cdot \delta$ if $\mathcal{P}$ is δ -worst-correct. One can also easily check that the running time of $\mathcal{B}$ can be bounded as $\text{Time}(\mathcal{B}) \approx 2 \cdot \text{Time}(\mathcal{A}) + O(q_H + q_G) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$. $\square$

Combining Lemma 8 and Theorem 6, we obtain the following lemma.

Lemma 10 (IND-CCA of $\text{KEM}_m^\perp$) Suppose $\mathcal{P}$ is δ -worst-correct and unique randomness recoverable.¹⁴ Let $\mathcal{A}$ be an IND-CCA adversary against $\text{KEM}_m^\perp$, making q_D , q_H and q_G queries to the decapsulation oracle, the random oracle H and the random oracle G , respectively. Let d_H (resp. d_G) be the query depth of $\mathcal{A}$'s H (resp. G) queries. Then there exists a OW-CPA adversary $\mathcal{B}$ against $\mathcal{P}$ such that

$$\begin{aligned} \text{Adv}_{\text{KEM}_m^\perp}^{\text{IND-CCA}}(\mathcal{A}) &\leq 8(d_G + q_D) \cdot \text{Adv}_{\mathcal{P}}^{\text{OW-CPA}}(\mathcal{B}) \\ &\quad + 20d_G \cdot \delta + O(q_H) \cdot \sqrt{\delta} + O(q_D) \cdot |\mathcal{R}|^{-1/2}. \end{aligned}$$

Let Rec be the recover algorithm of $\mathcal{P}$, the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx 2 \cdot \text{Time}(\mathcal{A}) + O(q_H + q_G) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec})) + O(q_H q_D + q_H^2)$.

As for $\text{KEM}_m^\perp$, we first introduce the following lemma, which shows that the security of $\text{KEM}_m^\perp$ implies the security $\text{KEM}_m^\perp$.

Lemma 11 (Explicit $\rightarrow$ implicit; (Bindel et al. 2019) Theorem 3) Let $\mathcal{P}$ be a rPKE. Let $\mathcal{A}$ be an IND-CCA adversary against $\text{KEM}_m^\perp$. Then there exists an IND-CCA adversary $\mathcal{B}$ against $\text{KEM}_m^\perp$, running in about the same time as $\mathcal{A}$, such that

$$\text{Adv}_{\text{KEM}_m^\perp, \mathcal{A}}^{\text{IND-CCA}} = \text{Adv}_{\text{KEM}_m^\perp, \mathcal{B}}^{\text{IND-CCA}}.$$

Hence, by the above lemma, our bound given in Corollary 10 also holds if we replace $\text{KEM}_m^\perp$ with $\text{KEM}_m^\perp$.

Appendix A: Proof of Lemma 4

Here, we give the detailed definition of quantum oracle algorithm $\mathcal{B}(z)$ with oracle access to H and O , where O can be G or $H \setminus S$. If we choose O to be G , $\mathcal{B}^{H, O}(z)$ defined here is the same as $\mathcal{B}^{H, G}(z)$ defined in Lemma 3. We also stress that the image set $f(S) := \{w^*\}$ is a single element.

Let $Q_{\mathcal{A}}$ denote the register containing all of $\mathcal{A}$'s state. Let B be a register defined over the set $\{0, 1\}$. Let W be a register defined over the set $\{0, 1\}^n$. Then the quantum oracle algorithm $\mathcal{B}^{H, O}(z)$, which acts on registers $(Q_{\mathcal{A}}, B, W)$, is defined as follows:

- Initialize registers (B, W) with $|0\rangle|0^n\rangle$.
- With input z , run the quantum oracle algorithm $\mathcal{A}^H(z)$.

¹⁴ Here we omit the condition of weakly γ -spreadness since if $\mathcal{P}$ is unique randomness recoverable, the value 2^γ equals $|\mathcal{R}|$ according to its definition.

- When $\mathcal{A}$ queries H with input/output register Q/R as part of Q_A , $\mathcal{B}$ answers this query by performing the following operations sequentially:

- Apply the unitary operation T_f on (Q, R, B, W) , which is defined as:

$$T_f|x, y, b, w\rangle := |x, y, b, w \oplus (b \cdot f(x))\rangle.$$

If $b = 1$, then $b \cdot f(x)$ equals to $f(x)$. Otherwise, it should be 0^n .

- Apply the following conditional operations according to the controlling bit in register B :
 1. If the state of B is $|+\rangle := (|0\rangle + |1\rangle)/\sqrt{2}$, query oracle H with input/output register Q/R .
 2. If the state of B is $|-\rangle := (|0\rangle - |1\rangle)/\sqrt{2}$, query oracle O with input/output register Q/R .
- Apply T_f on (Q, R, B, W) again.
- After $\mathcal{A}$ completes, $\mathcal{B}$ measures registers (B, W) , and suppose the measurement outcome is (b, w) . Then $\mathcal{B}$ outputs w if $b = 1$, outputs $\perp$ if $b = 0$.

Indeed, $\mathcal{B}$ invokes $\mathcal{A}$ in the black-box manner and only modifies the oracle calls of $\mathcal{A}$. The running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q) \cdot \text{Time}(f)$, since $\mathcal{B}$ needs to perform the unitary operation T_f with $2q$ times by definition. Now, we introduce the following lemma:

Lemma 12 *Let H, G, S, z be random variables sampled from some joint distribution. Let L be a register defined over the set $\{0, 1\}^q$. Let $\mathcal{C}^{H,S}(z)$ be an algorithm that operates on registers (Q_A, L) like $\mathcal{A}^H(z)$, except:*

- It initializes register L with $|0^q\rangle$.
- When $\mathcal{A}$ performs its i -th oracle queries on input/output register Q/R as part of Q_A , $\mathcal{C}$ first applies U_S on (Q, R, L) and then performs the oracle queries. Here U_S is defined as:

$$U_S|x, y, l\rangle := \begin{cases} |x, y, l\rangle & (x \notin S), \\ |x, y, \text{flip}_i(l)\rangle & (x \in S), \end{cases} \quad (\text{A1})$$

where the function $\text{flip}_i(l)$ flips the i -th bit of l .

Let ρ_{left} and ρ_{right} denote the final state of $\mathcal{A}^H(z)$ and $\mathcal{C}^{H,S}(z)$, respectively. Let P_{extract}' be the same as defined in Lemma 4. Then we have

$$B(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|, \rho_{\text{right}})^2 \leq 4P_{\text{extract}}'. \quad (\text{A2})$$

Here $B(\rho_1, \rho_2)$ is the Bures distance in Nielsen and Chuang (2016) of states ρ_1 and ρ_2 .

Proof We first construct a new algorithm based on $\mathcal{B}^{H,H \setminus S}(z)$. Let L be a register defined over the set $\{0, 1\}^q$. Let $\mathcal{D}^{H,S}(z)$ be an algorithm that operates on registers (Q_A, L, B, W) like $\mathcal{B}^{H,H \setminus S}(z)$, except:

- It initializes register L with $|0^q\rangle$.
- When $\mathcal{B}$ performs its i -th oracle queries to $H \setminus S$ on input/output register Q/R as part of Q_A , $\mathcal{D}$ first applies U_S on (Q, R, L) as defined in (A1) and then performs the oracle queries to H .

Indeed, $\mathcal{D}^{H,S}(z)$ is the 'deferred measurement' version of $\mathcal{B}^{H,H \setminus S}(z)$, and if we finally measure register L in the computational basis, the final state of $\mathcal{D}^{H,S}(z)$ is the same as that of $\mathcal{B}^{H,H \setminus S}(z)$. Then by the definition of P_{extract}' ,

$$P_{\text{extract}}' = \Pr[w \in f(S) : w \leftarrow \mathcal{D}^{H,S}(z)]. \quad (\text{A3})$$

Now we consider the situation when H, G, S, z are fixed. And without loss of generality, we suppose that $\mathcal{A}$ is unitary. If $\mathcal{A}$ is not unitary, we can use some auxiliary registers to construct a unitary variant of $\mathcal{A}$. Then, the joint state of $\mathcal{A}$ before the final measurement can be written as

$$|\Psi_{\mathcal{A}}\rangle := \prod_{i=1}^q (U_i \circ O_H) |\psi\rangle.$$

Here $|\psi\rangle$ is the initial pure state in register Q_A , $U_1, \dots, U_q$ are the unitary operations between the oracle query O_H , which maps $|x, y\rangle$ into $|x, y \oplus H(x)\rangle$.

According to the definition of $\mathcal{C}^{H,S}(z)$, its initial pure state should be $|\psi\rangle|0^q\rangle$, and its joint state before the final measurement can be written as

$$|\Psi_{\mathcal{C}}\rangle := \prod_{i=1}^q (U_i \circ O_H \circ U_S) |\psi\rangle|0^q\rangle.$$

Here $|0^q\rangle$ is the initial state in register L . Similarly, the initial pure state of $\mathcal{D}^{H,S}(z)$ is $|\psi\rangle|0^q\rangle|0,0^n\rangle$ and its joint state before the final measurement¹⁵ can be written as

$$|\Psi_{\mathcal{D}}\rangle := \prod_{i=1}^q (U_i \circ T_f \circ O_{H,S} \circ T_f) |\psi\rangle|0^q\rangle|0,0^n\rangle.$$

Here $|0^q\rangle|0,0^n\rangle$ is the initial state in registers (L, B, W) , and $O_{H,S}$ is a conditional operation defined as follows:

1. If the state of B is $|+\rangle := (|0\rangle + |1\rangle)/\sqrt{2}$, perform O_H .
2. If the state of B is $|-\rangle := (|0\rangle - |1\rangle)/\sqrt{2}$, perform $O_H \circ U_S$.

For $j \in \{1, 2, \dots, q\}$, let $|\Psi_{\mathcal{A}}^j\rangle$ be the internal state of $\mathcal{A}^H(z)$, written as

$$|\Psi_{\mathcal{A}}^j\rangle := \prod_{i=1}^j (U_i \circ O_H) |\psi\rangle.$$

And analogously for $|\Psi_{\mathcal{C}}^j\rangle$ and $|\Psi_{\mathcal{D}}^j\rangle$.

Then by using the induction, we get the fact that for $j \in \{1, 2, \dots, q\}$,

$$\begin{aligned} |\Psi_{\mathcal{D}}^j\rangle &= \frac{1}{2} \left(|\Psi_{\mathcal{A}}^j\rangle|0^q\rangle + |\Psi_{\mathcal{C}}^j\rangle \right) \otimes |0,0^n\rangle \\ &\quad + \frac{1}{2} \left(|\Psi_{\mathcal{A}}^j\rangle|0^q\rangle - |\Psi_{\mathcal{C}}^j\rangle \right) \otimes |1,w^*\rangle. \end{aligned}$$

That is to say, the joint state before the final measurement of $\mathcal{D}^{H,S}(z)$ can be rewritten as

$$\begin{aligned} |\Psi_{\mathcal{D}}\rangle &= \frac{1}{2} \left(|\Psi_{\mathcal{A}}\rangle|0^q\rangle + |\Psi_{\mathcal{C}}\rangle \right) \otimes |0,0^n\rangle \\ &\quad + \frac{1}{2} \left(|\Psi_{\mathcal{A}}\rangle|0^q\rangle - |\Psi_{\mathcal{C}}\rangle \right) \otimes |1,w^*\rangle. \end{aligned}$$

Hence, for fixed (H, G, S, z) , we have

$$\begin{aligned} \Pr[w \in f(S) : w \leftarrow \mathcal{D}^{H,S}(z), H, G, S, z] \\ = \frac{1}{4} \left\| |\Psi_{\mathcal{A}}\rangle|0^q\rangle - |\Psi_{\mathcal{C}}\rangle \right\|^2. \end{aligned} \quad (\text{A4})$$

Now we consider the situation when H, G, S, z are random, by the definitions of ρ_{left} and ρ_{right} , we have

$$\rho_{\text{left}} = \text{Exp}_{H,G,S,z} [|\Psi_{\mathcal{A}}\rangle\langle\Psi_{\mathcal{A}}|]$$

and

$$\rho_{\text{right}} = \text{Exp}_{H,G,S,z} [|\Psi_{\mathcal{C}}\rangle\langle\Psi_{\mathcal{C}}|].$$

Let $F(\rho_1, \rho_2)$ be the fidelity in Nielsen and Chuang (2016) of states ρ_1 and ρ_2 , then

$$\begin{aligned} &F(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|, \rho_{\text{right}}) \\ &= F \left(\text{Exp}_{H,G,S,z} [|\Psi_{\mathcal{A}}\rangle\langle\Psi_{\mathcal{A}}| \otimes |0^q\rangle\langle 0^q|], \text{Exp}_{H,G,S,z} [|\Psi_{\mathcal{C}}\rangle\langle\Psi_{\mathcal{C}}|] \right) \\ &\stackrel{(a)}{\geq} \text{Exp}_{H,G,S,z} [F(|\Psi_{\mathcal{A}}\rangle\langle\Psi_{\mathcal{A}}| \otimes |0^q\rangle\langle 0^q|, |\Psi_{\mathcal{C}}\rangle\langle\Psi_{\mathcal{C}}|)] \\ &\stackrel{(b)}{\geq} 1 - \frac{1}{2} \text{Exp}_{H,G,S,z} [\| |\Psi_{\mathcal{A}}\rangle|0^q\rangle - |\Psi_{\mathcal{C}}\rangle \|^2] \\ &\stackrel{(c)}{=} 1 - 2 \text{Exp}_{H,G,S,z} [\Pr[w \in f(S) : w \leftarrow \mathcal{D}^{H,S}(z), H, G, S, z]] \\ &= 1 - 2 [\Pr[w \in f(S) : w \leftarrow \mathcal{D}^{H,S}(z)]] \\ &\stackrel{(d)}{=} 1 - 2P_{\text{extract}}'. \end{aligned}$$

Here (a) follows the joint concavity in Nielsen and Chuang (2016) of the fidelity, (b) uses Lemma 3 in Ambainis et al. (2019), (c) and (d) use (A4) and (A3), respectively.

By definition of the Bures distance B that $B(\rho_1, \rho_2)^2 = 2(1 - F(\rho_1, \rho_2))$,

$$\begin{aligned} &B(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|, \rho_{\text{right}})^2 \\ &= 2(1 - F(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|, \rho_{\text{right}})) \leq 4P_{\text{extract}}', \end{aligned}$$

hence (A2) is proved. $\square$

Lemma 13 (Double-sided O2H, with semi-classical oracle–restated) *Let $H, G : X \rightarrow Y$ be random functions, z be a random value, and $S \subset X$ be a random set such that $\forall x \notin S, H(x) = G(x)$. (H, G, S, z may have arbitrary joint distribution.) Let $\mathcal{A}^H$ be a quantum oracle algorithm of query times q . Let $f : X \rightarrow W \subseteq \{0, 1\}^n$ be any function and $f(S)$ denote the image of S under f .*

We will define a quantum oracle algorithm $\mathcal{B}$ with access to H and G or H and $H \setminus S$ with running time $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q) \cdot \text{Time}(f)$, let

$$P_{\text{left}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^H(z)],$$

$$P_{\text{right}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^G(z)], \quad (\text{Case 1})$$

$$P_{\text{extract}} := \Pr[w \in f(S) : w \leftarrow \mathcal{B}^{H,G}(z)],$$

$$P_{\text{extract}}' := \Pr[w \in f(S) : w \leftarrow \mathcal{B}^{H,H \setminus S}(z)].$$

¹⁵ By the definition of $\mathcal{D}^{H,S}(z)$, its final measurement consists of two parts, including the final measurement of $\mathcal{A}^H(z)$ and the measurement on registers (B, W) .

If $f(S) = \{w^*\}$ is a single element, we have

$$|P_{\text{left}} - P_{\text{right}}| \leq 2\sqrt{P_{\text{extract}}}$$

and

$$|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}| \leq 2\sqrt{P_{\text{extract}}}.$$

The lemma also holds with the bound $2\sqrt{P_{\text{extract}}'}$ for the following alternative definitions of P_{right} :

$$P_{\text{right}} := \Pr[b = 1 : b \leftarrow \mathcal{A}^{H \setminus S}(z)], \quad (\text{Case 2})$$

$$P_{\text{right}} := \Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)], \quad (\text{Case 3})$$

$$P_{\text{right}} := \Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{G \setminus S}(z)], \quad (\text{Case 4})$$

$$P_{\text{right}} := \Pr[b = 1 \vee \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)], \quad (\text{Case 5})$$

$$P_{\text{right}} := \Pr[b = 1 \vee \text{Find} : b \leftarrow \mathcal{A}^{G \setminus S}(z)]. \quad (\text{Case 6})$$

Proof The definition of the quantum oracle algorithm $\mathcal{B}$ is given at the beginning of this section. Its running time is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q) \cdot \text{Time}(f)$. Moreover, as mentioned above, $\mathcal{B}^{H,G}$ has the same construction as “ $\mathcal{B}^{H,G}$ ” defined in Lemma 3. Therefore, Lemma 4 must hold for P_{right} in (case 1). So we only need to prove the other cases of P_{right} .

Let $P_{\mathbb{M}}(\rho)$ be the probability that $\mathbb{M}$ returns 1 when measuring a state ρ .

For the definition of P_{right} in (case 2), let $\mathbb{M}_{\mathcal{A}}$ be the measurement that measures what the output b of $\mathcal{A}$ is, then

$$\begin{aligned} |P_{\text{left}} - P_{\text{right}}| &= |P_{\mathbb{M}_{\mathcal{A}}}(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|) - P_{\mathbb{M}_{\mathcal{A}}}(\rho_{\text{right}})| \\ &\stackrel{(a)}{\leq} B(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|, \rho_{\text{right}}) \stackrel{(b)}{\leq} 4P_{\text{extract}}' \end{aligned}$$

Here (a) uses Lemma 4 in Ambainis et al. (2019), and (b) uses (A2). Similarly,

$$\begin{aligned} &|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}| \\ &= |\sqrt{P_{\mathbb{M}_{\mathcal{A}}}(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|)} - \sqrt{P_{\mathbb{M}_{\mathcal{A}}}(\rho_{\text{right}})}| \\ &\leq B(\rho_{\text{left}} \otimes |0^q\rangle\langle 0^q|, \rho_{\text{right}}) \\ &\leq 4P_{\text{extract}}' \end{aligned}$$

For P_{right} in (case 3), let $\mathbb{M}_{\mathcal{A}}^1$ be the measurement that measures whether $b = 1$ and L contains $|0^q\rangle$. Then by following the proof of (case 2), we can prove Lemma 4 for (case 3).

For P_{right} in (case 5), let $\mathbb{M}_{\mathcal{A}}^2$ be the measurement that measures whether $b = 1$ or L contains $|x\rangle$ where $x \neq 0^q$. Then by following the proof of (case 2), we can prove Lemma 4 for (case 5).

For P_{right} in (case 4), this follows immediately from (case 3), since Lemma 1 in Ambainis et al. (2019) indicates that

$$\begin{aligned} \Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)] \\ = \Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathcal{A}^{G \setminus S}(z)]. \end{aligned}$$

For P_{right} in (case 6), this follows immediately from (case 5), since Lemma 1 in Ambainis et al. (2019) also indicates that

$$\begin{aligned} \Pr[b = 1 \vee \text{Find} : b \leftarrow \mathcal{A}^{H \setminus S}(z)] \\ = \Pr[b = 1 \vee \text{Find} : b \leftarrow \mathcal{A}^{G \setminus S}(z)]. \end{aligned}$$

□

Appendix B: Cryptographic primitives

Definition 1 (Public Key Encryption) A public key encryption (PKE) scheme $\mathbf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ consists of three polynomial time algorithms (in the security parameter λ) with a finite message space $\mathcal{M}$ such that

1. **Gen**, the key generation algorithm, is a probabilistic algorithm that on input 1^λ outputs a key pair (pk, sk) .
2. **Enc**, the encryption algorithm, is a probabilistic algorithm that on input pk and a message $m \in \mathcal{M}$ outputs a ciphertext $c \leftarrow \text{Enc}_{pk}(m)$ that belongs to the ciphertext space $\mathcal{C}$. We also make the randomness used in encryption explicit by writing $c := \text{Enc}_{pk}(m; r)$, where $r \in \mathcal{R}$ ($\mathcal{R}$ is the randomness space).
3. **Dec**, the decryption algorithm, is a deterministic algorithm that on input sk and a ciphertext c outputs a message $m := \text{Dec}_{sk}(c)$ or a special symbol $\perp \notin \mathcal{M}$ to indicate that c is not a valid ciphertext.

We say $\mathbf{P}$ is a deterministic public key encryption (dPKE) scheme if **Enc** is a deterministic algorithm.

Definition 2 (Randomness Recoverability (Duman et al. 2023)) We say a PKE scheme $\mathbf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ is randomness recoverable if there exists a deterministic (recover) algorithm **Rec** such that for all $m \in \mathcal{M}$, $r \in \mathcal{R}$ and $c = \text{Enc}_{pk}(m; r)$,

GAME PRE-CPA _P ^A	GAME OW-CPA _P ^A	GAME POW-CPA _P ^A
1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $(pk, sk) \leftarrow \text{Gen}$
2 : $m^* \xleftarrow{\$} \mathcal{M}$	2 : $m^* \xleftarrow{\$} \mathcal{M}$	2 : $m_1^* \ m_2^* \xleftarrow{\$} \mathcal{M}_1 \times \mathcal{M}_2$
3 : $r^* \xleftarrow{\$} \mathcal{R}$	3 : $r^* \xleftarrow{\$} \mathcal{R}$	3 : $r^* \xleftarrow{\$} \mathcal{R}$
4 : $c^* := \text{Enc}_{pk}(m^*; r^*)$	4 : $c^* := \text{Enc}_{pk}(m^*; r^*)$	4 : $c^* := \text{Enc}_{pk}(m_1^* \ m_2^*; r^*)$
5 : $(m', r') \leftarrow \mathcal{A}(pk, c^*)$	5 : $m' \leftarrow \mathcal{A}(pk, c^*)$	5 : $m'_1 \leftarrow \mathcal{A}(pk, c^*, m_2^*)$
6 : return $[\text{Enc}_{pk}(m'; r') = c^*]$	6 : return $[m' = m^*]$	6 : return $[m'_1 = m_1^*]$

Fig. 8 Game PRE-CPA_P^A, OW-CPA_P^A, and POW-CPA_P^A. Here we suppose the message space of $\mathbb{P}$ is $\mathcal{M}_1 \times \mathcal{M}_2$ in game POW-CPA_P^A

$$\Pr \left[\begin{array}{l} \forall m' \in \text{Preimg}(pk, c), \\ \text{Enc}_{pk}(m'; \text{Rec}_{pk}(m', c)) \neq c \end{array} \right] = 0,$$

where $\text{Preimg}(pk, c) := \{m \in \mathcal{M} \mid \exists r \in \mathcal{R} : \text{Enc}_{pk}(m; r) = c\}$ and the probability above is taken over $(pk, sk) \leftarrow \text{Gen}$. Additionally, we require that $\text{Rec}_{pk}(m, c)$ returns the symbol $\perp \notin \mathcal{R}$ if $m \notin \text{Preimg}(pk, c)$.

Definition 3 (Unique Randomness Recoverability) We say a PKE scheme $\mathbb{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ is unique randomness recoverable if there exists a (recover) algorithm Rec such that for all $m \in \mathcal{M}$ and $r \in \mathcal{R}$,

$$\Pr[\text{Rec}_{pk}(m, c) \neq r \mid c := \text{Enc}_{pk}(m; r)] = 0,$$

where the probability is taken over $(pk, sk) \leftarrow \text{Gen}$. For the pair (m, c) , if $\nexists r \in \mathcal{R}$ such that $c = \text{Enc}_{pk}(m; r)$, $\text{Rec}_{pk}(m, c)$ returns the symbol $\perp \notin \mathcal{R}$.

Remark 1 Since Rec is deterministic, any PKE scheme satisfying the above equation implies that for $c := \text{Enc}_{pk}(m; r)$, $\nexists r' \neq r$ s.t. $c = \text{Enc}_{pk}(m; r')$. This is why we call this property as unique randomness recoverability.

Definition 4 (Correctness Error (Duman et al. 2023)) A PKE scheme $\mathbb{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ has worst (resp. average) correctness error δ if

$$\mathbb{E} \left[\max_{m \in \mathcal{M}} \Pr[\text{Dec}_{sk}(c) \neq m \mid c := \text{Enc}_{pk}(m; r), r \xleftarrow{\$} \mathcal{R}] \right]$$

(resp.

$$\mathbb{E} \left[\Pr \left[\text{Dec}_{sk}(c) \neq m \mid \begin{array}{l} c := \text{Enc}_{pk}(m; r), m \xleftarrow{\$} \mathcal{M} \\ r \xleftarrow{\$} \mathcal{R} \end{array} \right] \right]$$

) is less than or equal to δ . Here the expectation is taken over $(pk, sk) \leftarrow \text{Gen}$. We also say $\mathbb{P}$ is δ -worst-correct (resp. δ -average-correct).

Remark 2 From the above definition, it is obvious that δ -worst-correctness implies δ -average-correctness.

Definition 5 (Weakly γ -spread (Don et al. 2022)) A PKE scheme $\mathbb{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ is weakly γ -spread if

$$\mathbb{E} \left[\max_{m \in \mathcal{M}, c \in \mathcal{C}} \Pr[\text{Enc}_{pk}(m; r) = c \mid r \xleftarrow{\$} \mathcal{R}] \right]$$

is less than or equal to $2^{-\gamma}$. Here the expectation is taken over $(pk, sk) \leftarrow \text{Gen}$.

Definition 6 (Injectivity of a dPKE (Kuchta et al. 2020)) Let $\eta \geq 0$. A dPKE scheme $\mathbb{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ is η -injective if

$$\Pr[\text{Enc}_{pk}(\cdot) \text{ is not injective} \mid (pk, sk) \leftarrow \text{Gen}] \leq \eta.$$

If $\mathbb{P}$ is defined relative to a random oracle H , the Event “ $\text{Enc}_{pk}(\cdot)$ is not injective” should also be conditioned according to the event $H \xleftarrow{\$} \Omega_H$.

Definition 7 (Security notions for PKE) For any quantum adversary $\mathcal{A}$, we define its PRE-CPA, OW-CPA and POW-CPA advantage against PKE scheme $\mathbb{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ as follows:

$$\text{Adv}_{\mathbb{P}}^{\text{PRE-CPA}}(\mathcal{A}) := \Pr[1 \leftarrow \text{PRE-CPA}_{\mathbb{P}}^{\mathcal{A}}]$$

$$\text{Adv}_{\mathbb{P}}^{\text{OW-CPA}}(\mathcal{A}) := \Pr[1 \leftarrow \text{OW-CPA}_{\mathbb{P}}^{\mathcal{A}}]$$

$$\text{Adv}_{\mathbb{P}}^{\text{POW-CPA}}(\mathcal{A}) := \Pr[1 \leftarrow \text{POW-CPA}_{\mathbb{P}}^{\mathcal{A}}].$$

Here games PRE-CPA_P^A, OW-CPA_P^A and POW-CPA_P^A are shown in Fig. 8. We say $\mathbb{P}$ is PRE-CPA/OW-CPA/

GAME IND-CPA _{KEM} ^A	GAME IND-CCA _{KEM} ^A	oDeca(c)
1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $(pk, sk) \leftarrow \text{Gen}$	1 : if $c = c^*$
2 : $b \xleftarrow{\$} \{0, 1\}$	2 : $b \xleftarrow{\$} \{0, 1\}$	2 : return $\perp$
3 : $(K_0, c^*) \leftarrow \text{Enca}(pk)$	3 : $(K_0, c^*) \leftarrow \text{Enca}(pk)$	3 : else return $\text{Deca}(sk, c)$
4 : $K_1 \xleftarrow{\$} \mathcal{K}$	4 : $K_1 \xleftarrow{\$} \mathcal{K}$	
5 : $b' \leftarrow \mathcal{A}(pk, K_b, c^*)$	5 : $b' \leftarrow \mathcal{A}^{\text{oDeca}}(pk, K_b, c^*)$	
6 : return $\llbracket b' = b \rrbracket$	6 : return $\llbracket b' = b \rrbracket$	

Fig. 9 Game IND-CPA_{KEM}^A and IND-CCA_{KEM}^A. Random oracles involved in these games can be quantum accessed

POW-CPA secure, if for any quantum polynomial adversary $\mathcal{A}$, $\text{Adv}_{\mathbb{P}}^{\text{PRE-CPA}}(\mathcal{A})/\text{Adv}_{\mathbb{P}}^{\text{OW-CPA}}(\mathcal{A})/\text{Adv}_{\mathbb{P}}^{\text{POW-CPA}}(\mathcal{A})$ is negligible.

Definition 8 (Key Encapsulation Mechanism) A key encapsulation mechanism $\text{KEM} = (\text{Gen}, \text{Enca}, \text{Deca})$ consists of three polynomial algorithms with the key space $\mathcal{K}$ and the ciphertext space $\mathcal{C}$ such that

1. **Gen**, the key generation algorithm, is a probabilistic algorithm that on input 1^λ outputs a key pair (pk, sk) .
2. **Enca**, the encapsulation algorithm, is a probabilistic algorithm that on input pk outputs a key $k \in \mathcal{K}$ and a ciphertext $c \in \mathcal{C}$.
3. **Deca**, the decapsulation algorithm, is a deterministic algorithm that on input sk and a ciphertext c outputs outputs a key $K := \text{Deca}(sk, c)$ or a special symbol $\perp \notin \mathcal{K}$ to indicate that c is not a valid ciphertext.

Definition 9 (Security notions for KEM) For any quantum polynomial adversary $\mathcal{A}$, we define its IND-CPA and IND-CCA advantage against KEM scheme $\text{KEM} = (\text{Gen}, \text{Enca}, \text{Deca})$ as follows:

$$\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{A}) := \left| \Pr[1 \leftarrow \text{IND-CPA}_{\text{KEM}}^{\mathcal{A}}] - \frac{1}{2} \right|$$

$$\text{Adv}_{\text{KEM}}^{\text{IND-CCA}}(\mathcal{A}) := \left| \Pr[1 \leftarrow \text{IND-CCA}_{\text{KEM}}^{\mathcal{A}}] - \frac{1}{2} \right|.$$

Here game IND-CPA_{KEM}^A and game IND-CCA_{KEM}^A are shown in Fig. 9. We say KEM is IND-CPA (resp. IND-CCA) secure if for any quantum polynomial adversary $\mathcal{A}$, the advantage $\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{A})$ (resp. $\text{Adv}_{\text{KEM}}^{\text{IND-CCA}}(\mathcal{A})$) is negligible.

Appendix C: Omitted proofs in Sect. 4 and Sect. 5

C.1 Proof of Lemma 5

Proof For fixed (pk, sk) , we define the two following subsets of $\mathcal{M} \times \mathcal{R}$:

$$S_{pk,sk}^{\text{coll}} := \{(m, r) \mid \exists (m' \neq m, r') \in \mathcal{M} \times \mathcal{R},$$

$$\text{Enc}_{pk}(m'; r') = \text{Enc}_{pk}(m; r)\},$$

$$S_{pk,sk}^{\text{error}} := \{(m, r) \mid \text{Dec}_{sk}(\text{Enc}_{pk}(m; r)) \neq m\}.$$

According to the definition of δ -average-correctness, we have

$$\Pr[(m, r) \in S_{pk,sk}^{\text{error}} \mid (pk, sk) \leftarrow \text{Gen}, m \xleftarrow{\$} \mathcal{M}, r \xleftarrow{\$} \mathcal{R}] \leq \delta. \quad (\text{C7})$$

Denote the ciphertext space of $\mathbb{P}$ as $\mathcal{C}$. Then for $c \in \mathcal{C}$, define the set

$$S_{pk,sk,c}^{\text{coll}} := \{(m, r) : \text{Enc}_{pk}(m, r) = c\}.$$

By the definition of $S_{pk,sk}^{\text{coll}}$ given above, there must exist a subset $\mathcal{C} \subseteq \mathcal{C}$ such that $S_{pk,sk}^{\text{coll}} = \bigcup_{c \in \mathcal{C}} S_{pk,sk,c}^{\text{coll}}$. That is, we can separate $S_{pk,sk}^{\text{coll}}$ into several disjoint sets through the corresponding encryption value of $(m, r) \in S_{pk,sk}^{\text{coll}}$.

Since $\mathbb{P}$ is unique randomness recoverable, we state that for any $(m, r) \in \mathcal{M} \times \mathcal{R}$ and $(pk, sk) \leftarrow \text{Gen}$, there does not exist another different $r' \in \mathcal{R}$ such that $\text{Enc}_{pk}(m; r) = \text{Enc}_{pk}(m; r')$. Based on this uniqueness of the encryption randomness r , for $S_{pk,sk,c}^{\text{coll}}$, we can conclude that there exists at most one pair in $S_{pk,sk,c}^{\text{coll}}$ which does not belong to $S_{pk,sk}^{\text{error}}$ since the algorithm Dec is deterministic. And the other pairs in $S_{pk,sk,c}^{\text{coll}}$ must belong

GAMES $\mathbf{G_0-G_5}$		$H(x_H, y_H\rangle)$ (Let $r = \text{Rec}_{pk}(x_H M_2^*, c^*)$)	
1 : $(pk, sk) \leftarrow \text{Gen}$	// $\mathbf{G_0-G_5}$	1 : return $ x_H, y_H \oplus H(x_H)\rangle$	// $\mathbf{G_0-G_3}$
2 : $H \xleftarrow{\$} \Omega_H, M_1^* \xleftarrow{\$} \mathcal{M}_1$	// $\mathbf{G_0-G_5}$	2 : if $x_H = M_1^*$	// $\mathbf{G_4}$
3 : $u^* \xleftarrow{\$} \mathcal{M}_2, r^* \xleftarrow{\$} \mathcal{R}$	// $\mathbf{G_0-G_5}$	3 : return $ x_H, y_H \oplus u^*\rangle$	
4 : $m^* \xleftarrow{\$} \mathcal{M}$	// $\mathbf{G_0}$	4 : else return $ x_H, y_H \oplus H(x_H)\rangle$	
5 : $M_2^* := m^* \oplus H(M_1^*)$	// $\mathbf{G_0}$	5 : if $\text{Enc}_{pk}(x_H M_2^*; r) = c^*$	// $\mathbf{G_5}$
6 : $M_2^* \xleftarrow{\$} \mathcal{M}_2$	// $\mathbf{G_1-G_5}$	6 : return $ x_H, y_H \oplus u^*\rangle$	
7 : $m^* := M_2^* \oplus H(M_1^*)$	// $\mathbf{G_1-G_3}$	7 : else return $ x_H, y_H \oplus H(x_H)\rangle$	
8 : $m^* := M_2^* \oplus u^*$	// $\mathbf{G_4-G_5}$	$\mathcal{O}_{\{M_1^*\}}^{SC}(x_H, y_H\rangle)$	
9 : $c^* := \text{Enc}_{pk}(M_1^* M_2^*; r^*)$	// $\mathbf{G_0-G_5}$	1 : initialize register B with $ 0\rangle$	// $\mathbf{G_2-G_5}$
10 : $m' \leftarrow \mathcal{A}^H(pk, c^*)$	// $\mathbf{G_0-G_1}$	2 : perform	// $\mathbf{G_2-G_4}$
11 : $m' \leftarrow \mathcal{A}^{H \setminus \{M_1^*\}}(pk, c^*)$	// $\mathbf{G_2}$	3 : $ x_H, y_H\rangle 0\rangle \mapsto x_H, y_H\rangle f_{M_1^*}(x_H)\rangle$	
12 : $M_1' \leftarrow \mathcal{A}_1^{H, H \setminus \{M_1^*\}}(pk, c^*)$	// $\mathbf{G_3-G_5}$	4 : perform	// $\mathbf{G_5}$
13 : return $[m' = m^*]$	// $\mathbf{G_0-G_2}$	5 : $ x_H, y_H\rangle 0\rangle \mapsto x_H, y_H\rangle f'_{M_1^*}(x_H)\rangle$	
14 : return $[M_1' = M_1^*]$	// $\mathbf{G_3-G_5}$	6 : measure register B , return the result	// $\mathbf{G_2-G_5}$

Fig. 10 Games $\mathbf{G_0}$ to $\mathbf{G_5}$ in the proof of Theorem 3. The oracles H and $\mathcal{O}_{\{M_1^*\}}^{SC}$ can both be queried in the superposition, while in this figure we just write the input state of H and $\mathcal{O}_{\{M_1^*\}}^{SC}$ as $|x_H, y_H\rangle$ for brevity

to $S_{pk,sk,c}^{coll} \cap S_{pk,sk}^{error}$. Note that $|S_{pk,sk,c}^{coll}| \geq 2$ for any $c \in C$, it is easy to check that

$$|S_{pk,sk,c}^{coll} \setminus S_{pk,sk}^{error}| \leq |S_{pk,sk,c}^{coll} \cap S_{pk,sk}^{error}|.$$

Since $S_{pk,sk}^{coll} = \bigcup_{c \in C} S_{pk,sk,c}^{coll}$ and $S_{pk,sk,c}^{coll}$ are disjoint for different $c \in C$, we have

$$|S_{pk,sk}^{coll} \setminus S_{pk,sk}^{error}| \leq |S_{pk,sk}^{coll} \cap S_{pk,sk}^{error}|. \quad (\text{C8})$$

Then, for $(pk, sk) \leftarrow \text{Gen}$, $m \xleftarrow{\$} \mathcal{M}$, and $r \xleftarrow{\$} \mathcal{R}$, we get

$$\begin{aligned} & \Pr \left[\begin{array}{l} \exists (m' \neq m, r') \in \mathcal{M} \times \mathcal{R}, \\ \text{Enc}_{pk}(m'; r') = \text{Enc}_{pk}(m; r) \end{array} \right] \\ &= \Pr[(m, r) \in S_{pk,sk}^{coll}] + \Pr[(m, r) \in S_{pk,sk}^{coll} \cap S_{pk,sk}^{error}] \\ &\stackrel{(a)}{\leq} 2 \Pr[(m, r) \in S_{pk,sk}^{coll} \cap S_{pk,sk}^{error}] \\ &\leq 2 \Pr[(m, r) \in S_{pk,sk}^{error}] \\ &\stackrel{(b)}{\leq} 2\delta. \end{aligned}$$

Here (a) and (b) use (C8) and (C7), respectively.

By the definition of correctness error given in Appendix B, the δ -worst-correctness must imply the δ

-average-correctness. Hence, the above proof also holds for a PKE scheme $\mathbf{P}$ that is δ -worst-correct. $\square$

C.2 Proof of Theorem 3

Proof To prove this theorem, we define a sequence of games (see also Fig. 10).

Game $\mathbf{G_0}$: This is the OW-CPA game of ACWC[P, H] with adversary $\mathcal{A}$.

$$\text{Adv}_{\text{ACWC}[P,H]}^{\text{OW-CPA}}(\mathcal{A}) = \Pr[1 \leftarrow \mathbf{G_0}]. \quad (\text{C9})$$

Game $\mathbf{G_1}$: The only difference between this game and game $\mathbf{G_0}$ is the way of getting m^* and M_2^* . In this game, M_2^* is uniformly sampled from $\mathcal{M}_2$ at first, and then m^* is computed as $M_2^* \oplus H(M_1^*)$.

$$\Pr[1 \leftarrow \mathbf{G_0}] = \Pr[1 \leftarrow \mathbf{G_1}]. \quad (\text{C10})$$

Game $\mathbf{G_2}$: Based on game $\mathbf{G_1}$, the semi-classical oracle $\mathcal{O}_{\{M_1^*\}}^{SC}$ is introduced and the oracle H is changed into $H \setminus \{M_1^*\}$. Here $\mathcal{O}_{\{M_1^*\}}^{SC}$ is simulated by using the function

$$f_{M_1^*}(x_H) := \begin{cases} 1 & \text{if } x_H = M_1^*, \\ 0 & \text{if } x_H \neq M_1^*. \end{cases}$$

By using Lemma 4, where $S := \{M_1^*\}$ and $f : \mathcal{M}_1 \rightarrow \mathcal{M}_1$ is an identical function, there exists an adversary $\mathcal{A}_1$ with quantum oracle access to H and $H \setminus \{M_1^*\}$ such that

$$\begin{aligned} & \left| \sqrt{\Pr[1 \leftarrow \mathbf{G}_1]} - \sqrt{\Pr[b \leftarrow \mathbf{G}_2 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_2]} \right| \\ & \leq 2\sqrt{\Pr[M_1' = M_1^* : M_1' \leftarrow \mathcal{A}_1^{H, H \setminus \{M_1^*\}}(pk, c^*)]}. \end{aligned} \quad (\text{C11})$$

Here (pk, c^*) is generated in the same way as game $\mathbf{G}_2$, Find is the event that the semi-classical oracle $\mathcal{O}_{\{M_1^*\}}^{SC}$ ever returns 1 in game $\mathbf{G}_2$, and the running time of $\mathcal{A}_1$ is almost the same as $\mathcal{A}$.

Similar to (12) in the proof of Theorem 2, we also have

$$\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_2] \leq \frac{1}{|\mathcal{M}|}. \quad (\text{C12})$$

Game $\mathbf{G}_3$: This game is the same as game $\mathbf{G}_2$, except $\mathcal{A}$ is changed into $\mathcal{A}_1$.

$$\begin{aligned} & \Pr[1 \leftarrow \mathbf{G}_3] \\ & = \Pr[M_1' = M_1^* : M_1' \leftarrow \mathcal{A}_1^{H, H \setminus \{M_1^*\}}(pk, c^*)]. \end{aligned} \quad (\text{C13})$$

Game $\mathbf{G}_4$: In this game, $H(M_1^*)$ is reprogrammed into u^* , where $u^* \xleftarrow{\$} \mathcal{M}$ is sampled at the beginning of the game.

$$\Pr[1 \leftarrow \mathbf{G}_3] = \Pr[1 \leftarrow \mathbf{G}_4]. \quad (\text{C14})$$

Game $\mathbf{G}_5$: This game is the same as game $\mathbf{G}_4$, except that the simulation of oracles H and $\mathcal{O}_{\{M_1^*\}}^{SC}$ is changed as following:

1. Simulation of H : If the input x_H satisfies $\text{Enc}_{pk}(x_H || M_2^*; \text{Rec}_{pk}(x_H || M_2^*, c^*))$ equals to c^* , then u^* will be returned. Otherwise, $H(x_H)$ is returned.
2. Simulation of $\mathcal{O}_{\{M_1^*\}}^{SC}$: The function $f_{M_1^*}$ is changed into

$$\begin{aligned} & f_{M_1^*}'(x_H) := \\ & \begin{cases} 1 & \text{if } \text{Enc}_{pk}(x_H || M_2^*; \text{Rec}_{pk}(x_H || M_2^*, c^*)) = c^*, \\ 0 & \text{if } \text{Enc}_{pk}(x_H || M_2^*; \text{Rec}_{pk}(x_H || M_2^*, c^*)) \neq c^*. \end{cases} \end{aligned}$$

Let Bad the event that

$\exists (M_1 \neq M_1^*, r) \in (\mathcal{M} \times \mathcal{R})$ such that

$\text{Enc}_{pk}(M_1 || M_2^*; r) = c^*$.

If Bad does not occur in game $\mathbf{G}_5$, then $\text{Enc}_{pk}(x_H || M_2^*; \text{Rec}_{pk}(x_H || M_2^*, c^*))$ equals to c^* iff x_H

equals to M_1^* . This means that game $\mathbf{G}_4$ and game $\mathbf{G}_5$ are exactly the same if Bad does not occur. Thus by using the difference lemma in Shoup (2004),

$$|\Pr[1 \leftarrow \mathbf{G}_4] - \Pr[1 \leftarrow \mathbf{G}_5]| \leq \Pr[\text{Bad}]. \quad (\text{C15})$$

Since $\mathbf{P}$ is δ -average-correct, by using Lemma 5, we obtain

$$\Pr[\text{Bad}] \leq 2 \cdot \delta. \quad (\text{C16})$$

Now we define a POW-CPA adversary $\mathcal{B}$ against $\mathbf{P}$ as follows:

1. The input of adversary $\mathcal{B}$ is the $(pk, c^* := \text{Enc}_{pk}(M_1^* || M_2^*; r^*), M_2^*)$, where $M_1^* \xleftarrow{\$} \mathcal{M}_1, M_2^* \xleftarrow{\$} \mathcal{M}_2$ and $r^* \xleftarrow{\$} \mathcal{R}$.
2. According to the method provided in Sect. 2.2, we choose a function H to simulate a QRO with input/output space $\mathcal{M}_1/\mathcal{M}_2$.
3. Choose $u^* \xleftarrow{\$} \mathcal{M}_2$, run the adversary $\mathcal{A}_1$ with input (pk, c^*) in game $\mathbf{G}_5$. When $\mathcal{A}_1$ queries H or $H \setminus \{M_1^*\}$, answer it in the same way as in game $\mathbf{G}_5$.

Note that $f_{M_1^*}'$ can be computed by using the algorithms Enc and Rec , so the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$. Then

$$\text{Adv}_{\mathbf{P}}^{\text{POW-CPA}}(\mathcal{B}) = \Pr[1 \leftarrow \mathbf{G}_5]. \quad (\text{C17})$$

Combining (C9) to (C17), we get

$$\text{Adv}_{\text{ACWC}[\mathbf{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) \leq 8 \cdot \text{Adv}_{\mathbf{P}}^{\text{POW-CPA}}(\mathcal{B}) + 16 \cdot \delta + \frac{2}{|\mathcal{M}|}.$$

□

C.3 Proof of Lemma 6

Here we prove Lemma 6 by proving the following two claims.

Claim 1: For unique randomness recoverable PKE scheme $\mathbf{P}$, then so is $\text{POTO}[\mathbf{P}, G]$.

Proof According to the unique randomness recoverability of $\mathbf{P}$, there exists a recover algorithm Rec for $\mathbf{P}$ such that for all $m_1 \in \mathcal{M}$ and $r \in \mathcal{R}_1 \times \mathcal{R}_2$,

$$\Pr[\text{Rec}_{pk}(m_1, \text{Enc}_{pk}(m_1; r)) \neq r] = 0.$$

Here the probability is taken over $(pk, sk) \leftarrow \text{Gen}$.

Now we define the recover algorithm Rec' for $\text{POTO}[\mathbf{P}, G]$ as follows:

1. The input is $(pk, m_1 || m_2, c)$.
2. Run $\text{Rec}_{pk}(m_1, c)$ to get its output r . If $r = \perp$, then return $\perp$. Otherwise, parse r as $r_1 || r_2 \in \mathcal{R}_1 \times \mathcal{R}_2$ and return r_1 .

It is not hard to check that for all $m_1 || m_2 \in \mathcal{M} \times \mathcal{R}_2$ and $r_1 \in \mathcal{R}_1$,

$$\Pr[\text{Rec}_{pk}'(m_1 || m_2, \text{Enc}_{pk}'(m_1 || m_2; r_1)) \neq r_1] = 0,$$

where the probability is taken over $(pk, sk) \leftarrow \text{Gen}$. $\square$

Claim 2: If unique randomness recoverable $\mathbf{P}$ is δ -average-correct, then $\text{POTO}[\mathbf{P}, G]$ is also δ -average-correct.

Proof For $(pk, sk) \leftarrow \text{Gen}$, $r \xleftarrow{\$} \mathcal{R}_1 \times \mathcal{R}_2$, and $m \xleftarrow{\$} \mathcal{M}$, let Fail be the event that $\text{Dec}_{sk}(\text{Enc}_{pk}(m; r)) \neq m$. Since $\mathbf{P}$ is δ -average-correct, we have $\Pr[\text{Fail}] \leq \delta$.

For $(pk, sk) \leftarrow \text{Gen}$, $r_1 \xleftarrow{\$} \mathcal{R}_1$, $G \xleftarrow{\$} \Omega_G$, and $m_1 || m_2 \xleftarrow{\$} \mathcal{M} \times \mathcal{R}_2$, by the construction of $\text{POTO}[\mathbf{P}, G]$ given in Fig. 5 and the unique randomness recoverability of $\mathbf{P}$, it is not hard to check that

$$\Pr[\text{Dec}_{sk}'(\text{Enc}_{pk}'(m_1 || m_2; r_1)) \neq m_1 || m_2 \mid \neg \text{Fail}] = 0.$$

Then, we have

$$\begin{aligned} & \Pr[\text{Dec}_{sk}'(\text{Enc}_{pk}'(m_1 || m_2; r_1)) \neq m_1 || m_2] \\ & \leq \Pr[\text{Fail}] + \Pr[\text{Dec}_{sk}'(\text{Enc}_{pk}'(m_1 || m_2; r_1)) \\ & \quad \neq m_1 || m_2 \mid \neg \text{Fail}] \\ & = \Pr[\text{Fail}] \leq \delta. \end{aligned}$$

This means that the PKE scheme $\text{POTO}[\mathbf{P}, G]$ is also δ -average-correct. $\square$

C.4 Proof of Lemma 7

Transformation T: Let $\mathbf{P} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a PKE scheme with message space $\mathcal{M}$ and randomness space $\mathcal{R}$. Let $H : \mathcal{M} \rightarrow \mathcal{R}$ be a hash function. We associate $\mathbf{T}[\mathbf{P}, H] := (\text{Gen}, \text{Enc}', \text{Dec}')$. The constituting algorithms of $\mathbf{T}[\mathbf{P}, H]$ are given in Fig. 11.

Proof To prove this theorem, we define a sequence of games (see also Fig. 12).

Game $\mathbf{G}_0$: This is the OW-CPA game of $\mathbf{T}[\mathbf{P}, H]$ with $\mathcal{A}$.

$$\text{Adv}_{\mathbf{T}[\mathbf{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) = \Pr[1 \leftarrow \mathbf{G}_0]. \quad (\text{C18})$$

Game $\mathbf{G}_1$: This game is the same as game $\mathbf{G}_0$, except that $\mathcal{A}$ is changed into $\mathcal{A}_1$. Here $\mathcal{A}_1$ first runs $\mathcal{A}$ to get the output m' , then it makes an additional query to H on m' and ignores its result. Finally it outputs m' .

$$\Pr[1 \leftarrow \mathbf{G}_1] = \Pr[1 \leftarrow \mathbf{G}_0]. \quad (\text{C19})$$

Game $\mathbf{G}_2$: In this game, the oracle H is changed into $H \setminus \{m^*\}$. Here $H \setminus \{m^*\}$ is an oracle that first queries the semi-classical oracle $\mathcal{O}_{\{m^*\}}^{\text{SC}}$ and then H . The oracle $\mathcal{O}_{\{m^*\}}^{\text{SC}}$ is simulated by using the function

$$f_{m^*}(x_H) := \begin{cases} 1 & \text{if } x_H = m^*, \\ 0 & \text{if } x_H \neq m^*. \end{cases}$$

Let f be the identical function that $f(x) = x$ for any x , then by using (case 3) of our Lemma 4 with $S := \{m^*\}$ and f , there exists an adversary $\mathcal{A}_2$ with quantum oracle access to H and $H \setminus \{m^*\}$ such that

$$\begin{aligned} & \left| \sqrt{\Pr[1 \leftarrow \mathbf{G}_1]} - \sqrt{\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow \mathbf{G}_2]} \right| \\ & \leq 2\sqrt{\Pr[m' = m^* : m' \leftarrow \mathcal{A}_2^{H, H \setminus \{m^*\}}(pk, c^*)]}. \end{aligned} \quad (\text{C20})$$

Gen	$\text{Enc}_{pk}'(m \in \mathcal{M})$	$\text{Dec}_{sk}'(c)$
1 : $(pk, sk) \leftarrow \text{Gen}$	1 : $c := \text{Enc}_{pk}(m; H(m))$	1 : $m' := \text{Dec}_{sk}(c)$
2 : return (pk, sk)	2 : return c	2 : if $m' = \perp$ or $c \neq \text{Enc}_{pk}(m'; H(m'))$
		3 : return $\perp$
		4 : else return m'

Fig. 11 Public Key Encryption $\mathbf{T}[\mathbf{P}, H] = (\text{Gen}, \text{Enc}', \text{Dec}')$

GAMES G_0 - G_5		$H(x_H, y_H\rangle)$
1 : $(pk, sk) \leftarrow \text{Gen}$	// G_0 - G_5	1 : return $ x_H, y_H \oplus H(x_H)\rangle$ // G_0 - G_3
2 : $H \xleftarrow{\$} \Omega_H$	// G_0 - G_3	2 : if $x_H = m^*$ // G_4
3 : $H_1 \xleftarrow{\$} \Omega_H$	// G_4 - G_5	3 : return $ x_H, y_H \oplus \text{Rec}_{pk}(x_H, c^*)\rangle$
4 : $r^* \xleftarrow{\$} \mathcal{R}$	// G_4 - G_5	4 : else return $ x_H, y_H \oplus H_1(x_H)\rangle$
5 : $m^* \xleftarrow{\$} \mathcal{M}$	// G_0 - G_5	5 : if $\text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c^*)) = c^*$ // G_5
6 : $c^* := \text{Enc}_{pk}(m^*; H(m^*))$	// G_0 - G_3	6 : return $ x_H, y_H \oplus \text{Rec}_{pk}(x_H, c^*)\rangle$
7 : $c^* := \text{Enc}_{pk}(m^*; r^*)$	// G_4 - G_5	7 : else return $ x_H, y_H \oplus H_1(x_H)\rangle$
8 : $m' \leftarrow \mathcal{A}_1^H(pk, c^*)$	// G_0	$\mathcal{O}_{\{m^*\}}^{SC}(x_H, y_H\rangle)$
9 : $m' \leftarrow \mathcal{A}_1^H(pk, c^*)$	// G_1	1 : initialize register B with $ 0\rangle$ // G_2 - G_5
10 : $m' \leftarrow \mathcal{A}_1^{H \setminus \{m^*\}}(pk, c^*)$	// G_2	2 : perform // G_2 - G_4
11 : $m' \leftarrow \mathcal{A}_2^{H, H \setminus \{m^*\}}(pk, c^*)$	// G_3 - G_5	3 : $ x_H, y_H\rangle 0\rangle \mapsto x_H, y_H\rangle f_{m^*}(x_H)\rangle$
12 : return $[m' = m^*]$	// G_0 - G_5	4 : perform // G_5
		5 : $ x_H, y_H\rangle 0\rangle \mapsto x_H, y_H\rangle f'_{m^*}(x_H)\rangle$
		6 : measure register B , return the result // G_2 - G_5

Fig. 12 Games G_0 to G_5 in the proof of Lemma 7. The oracles H and $\mathcal{O}_{\{m^*\}}^{SC}$ can both be queried in the superposition, while in this figure we just write the input state of H and $\mathcal{O}_{\{m^*\}}^{SC}$ as $|x_H, y_H\rangle$ for brevity

Here (pk, c^*) is generated in the same way as game G_2 , Find is the event that the semi-classical oracle $\mathcal{O}_{\{m^*\}}^{SC}$ ever returns 1 in game G_2 .

According to the definition of $\mathcal{A}_1$, if the final output m' equals m^* , then the event Find must have occurred in game G_2 . Hence, we have

$$\Pr[b = 1 \wedge \neg \text{Find} : b \leftarrow G_2] = 0. \quad (C21)$$

Game G_3 : This game is the same as game G_2 , except that $\mathcal{A}_1$ is changed into $\mathcal{A}_2$. Here $\mathcal{A}_2$ has oracle access to H and $H \setminus \{m^*\}$.

$$\begin{aligned} \Pr[m' = m^* : m' \leftarrow \mathcal{A}_2^{H, H \setminus \{m^*\}}(pk, c^*)] \\ = \Pr[1 \leftarrow G_3]. \end{aligned} \quad (C22)$$

Game G_4 : In this game, we add a new random oracle $H_1 : \mathcal{M} \rightarrow \mathcal{R}$, and $H(m^*)$ is replaced by $r^* \xleftarrow{\$} \mathcal{R}$. The simulation of H is also changed, it outputs $\text{Rec}_{pk}(x_H, c^*)$ (resp. $H_1(x)$) for input $x_H = m^*$ (resp. $x_H \neq m^*$).

By the definition of unique randomness recoverability given in Appendix B, the r^* , m^* and c^* in game G_4 must satisfy $\text{Rec}_{pk}(m^*, c^*) = r^*$. Thus, we have

$$\Pr[1 \leftarrow G_3] = \Pr[1 \leftarrow G_4]. \quad (C23)$$

Game G_5 : This game is the same as game G_4 , except that the simulation of oracles H and $\mathcal{O}_{\{m^*\}}^{SC}$ is changed as following:

1. Simulation of H : If the input x_H satisfies $\text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c^*)) = c^*$, then $\text{Rec}_{pk}(x_H, c^*)$ will be returned. Otherwise, $H_1(x_H)$ is returned.
2. Simulation of $\mathcal{O}_{\{m^*\}}^{SC}$: The function f_{m^*} is changed into

$$f_{m^*}'(x_H) := \begin{cases} 1 & \text{if } \text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c^*)) = c^*, \\ 0 & \text{if } \text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c^*)) \neq c^*. \end{cases}$$

Let Bad be the event that there exists $m \neq m^*$ and $r \in \mathcal{R}$ such that $\text{Enc}_{pk}(m; r) = c^*$. When Bad does not occur, $\text{Enc}_{pk}(x_H; \text{Rec}_{pk}(x_H, c^*)) = c^*$ iff $x_H = m^*$. Then by the difference lemma in Shoup (2004),

$$|\Pr[1 \leftarrow G_4] - \Pr[1 \leftarrow G_5]| \leq \Pr[\text{Bad}]. \quad (C24)$$

Since PKE scheme $\mathcal{P}$ is δ -average-correct, by using our Lemma 5, we get

$$\Pr[\text{Bad}] \leq 2 \cdot \delta. \quad (C25)$$

Now we define a OW-CPA adversary $\mathcal{B}$ against $\mathcal{P}$ as follows:

1. The input is $(pk, c^* := \text{Enc}_{pk}(m^*; r^*))$, where $m^* \xleftarrow{\$} \mathcal{M}$ and $r^* \xleftarrow{\$} \mathcal{R}$.
2. According to the method provided in Sect. 2.2, we choose a function H_1 to simulate a QRO with input/output space $\mathcal{M}/\mathcal{R}$.
3. Run the adversary $\mathcal{A}_2$ in game $\mathbf{G}_5$ with the input (pk, c^*) . When $\mathcal{A}_2$ queries H or $H \setminus \{m^*\}$, answer it in the same way as game $\mathbf{G}_5$.

Note that f_{m^*}' can be computed by using the algorithms Enc and Rec, the running time of $\mathcal{B}$ is $\text{Time}(\mathcal{B}) \approx \text{Time}(\mathcal{A}) + O(q_H) \cdot (t_{\text{RO}} + \text{Time}(\text{Enc}) + \text{Time}(\text{Rec}))$, and we have

$$\text{Adv}_{\mathcal{P}}^{\text{OW-CPA}}(\mathcal{B}) = \Pr[1 \leftarrow \mathbf{G}_5]. \quad (\text{C26})$$

Combining (C18) to (C26), we finally obtain

$$\text{Adv}_{\mathcal{T}[\mathcal{P}, H]}^{\text{OW-CPA}}(\mathcal{A}) \leq 4 \cdot \text{Adv}_{\mathcal{P}}^{\text{OW-CPA}}(\mathcal{B}) + 8 \cdot \delta.$$

By the definition of correctness error given in Appendix B, the δ -worst-correctness must imply the δ -average-correctness. Hence, the above proof also holds for a PKE scheme $\mathcal{P}$ that is δ -worst-correct. $\square$

Acknowledgements

We would like to thank the anonymous reviewers of Cybersecurity for their insightful comments and suggestions.

Author contributions

Jiawei Bao, Jiangxia Ge and Rui Xue participated in problem discussions and manuscript writing process. All authors read and approved the final manuscript.

Funding

This work was supported by National Natural Science Foundation of China (Grants No.62172405).

Availability of data and materials

Not applicable.

Declarations

Conflict of interest

The authors declare that they have no competing interests.

Received: 11 January 2024 Accepted: 29 February 2024

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Publisher's Note

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